User stories: FaceAttend EDU

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Planned stories (pending documentation)

These stories are cited by other user stories and tracked in the traceability matrix (fae-docs), but their detailed specification is not yet part of this annex.

Table 1

Planned stories (pending documentation)

ID	Title	Requirement	Sprint	Cited in
HU-ATT-005	Alternative registration for NN users	RF 3.2.1	Sprint 2	HU-ATT-003, HU-ATT-004
HU-IAM-004	Massive user import via CSV	RF 1.2	Sprint 2	HU-IAM-002
HU-IAM-007	Activate/deactivate users	RF 1.6	Sprint 2	HU-CONF-006, HU-CONF-007, HU-HIST-002
HU-IAM-010	Change user role	RF 4.6	Sprint 2	HU-HIST-002

Source. Own elaboration.

Note: To navigate to a specific story, click the link for the corresponding module or story.

HU-IAM

HU-IAM-001: User login

Story

As a registered user (student, instructor, administrator)

I want to log in to FaceAttend EDU using my credentials

So that I can securely access the modules and information allowed by my role.

Acceptance criteria

Format: “Given [context/initial state], when [user action], then [expected and verifiable result].”

- [ ] *AC1:* Given that a user is registered and active in the system, when the user submits valid credentials (`identifier` = email or username, plus password), then the system authenticates the user, creates an *opaque session* (`identity.user_session` row returning a `sessionId` used as `Authorization: Bearer`), and grants access according to the assigned role.
- [ ] *AC2:* Given that a user submits invalid credentials, when the login request is processed, then the system denies access with `401` and the generic message *`Invalid email or password`*, without revealing whether the identifier exists.
- [ ] *AC3:* Given that a user account is inactive or deactivated, when the user attempts to log in, then the system denies access with the same generic `401 Invalid email or password` (implementation note: a dedicated "account not active" message is *not* returned — deliberate, to avoid account enumeration; `AuthenticateUserUseCaseImpl`).
- [ ] *AC4:* Given that a user repeatedly submits invalid credentials, when the configured failed-attempt threshold is reached, then the system temporarily blocks or delays further attempts and records the event for auditing.

  Status: not implemented — `configuration.security_configuration` seeds `max_login_attempts=5` and `lockout_duration_minutes=30`, but no

service reads them yet. Recorded as pending work.

- [ ] *AC5:* Given that the system is running in a production or pilot environment, when the user submits login credentials, then the communication is protected using HTTPS.
- [ ] *AC6:* Given normal network conditions, when the user submits valid credentials, then the login process responds in less than 5 seconds, according to RNF 1.
- [ ] *AC7:* Given that the login is successful, when the user accesses the system, then the system records a login audit event with date, time, user identifier, and result.

  Implemented as platform logging + `last_access` update on `app_user` (the Audit bounded context was removed, 2026-09).

Technical notes

- The SRS RF 1.3 presents an inconsistency: the input is described as "Registro en FaceAttend EDU", but the correct input is the *`identifier` field* (email *or* username — `@JsonAlias({"username","email"})`) plus password.
- Passwords must not be stored or transmitted in plain text; hashing is *bcrypt cost 12, `$2b`* (ADR-008, `SecurityConfig`).
- Failed login attempts are logged without exposing sensitive data (generic 401 always).
- There is *no JWT and no separate access/refresh token pair*: the token is the `sessionId` (opaque UUID) of a row in `identity.user_session` (`session_status` Active/Closed), valid for 480 minutes. `POST /api/v1/auth/refresh` rotates the session (single use) and `POST /api/v1/auth/logout?sessionId=<UUID>` closes it. Kong has no `jwt` plugin: a *pre-function* validates the session against identity, and each service validates it again (`AuthTokenFilter` in Java, `authGuard.ts` in TypeScript, `security.py` in Python).
- The login flow must consider role-based redirection after successful authentication (Web maps roles via `backendRoles.js`: `Administrador|Instructor|Aprendiz`, legacy names accepted).

- Sensitive authentication data must not be written to logs or shown in the user interface.

Responsible service(s): `ms-identity` (authentication is merged into Identity; there is no separate IAM service).

Endpoint(s) implemented:

- `POST /api/v1/auth/login` — implemented (`AuthController`).
- `POST /api/v1/auth/logout` — implemented.
- `GET /api/v1/auth/me` — implemented (returns the full `UserDto`).

Events generated:

- `UserLoggedIn`
- `LoginFailed`
- `UserTemporarilyBlocked`

Required permissions:

- Public endpoint for login.
- Authenticated session required after successful login.

Definition of Done (DoD)

This HU can only be closed when it meets the team's full DoD.

Additional checks specific to this HU:

- [ ] Valid login scenario tested.
- [ ] Invalid login scenario tested.
- [ ] Inactive user scenario tested.
- [ ] Repeated failed attempts scenario tested.
- [ ] Password is not exposed in logs, responses, or UI.
- [ ] HTTPS is enforced in production/pilot.
- [ ] Login audit event is generated.
- [ ] Error messages are clear but do not reveal sensitive information.

- [ ] Login response time is validated under normal conditions and remains under 5 seconds.
- [ ] Traceability to RF 1.3 is documented.

Estimation and priority

Table 2

Estimation and priority

Field	Value
Story Points	5
Priority	High
Target sprint	Sprint 1 / MVP
Dependencies	HU-IAM-002: User registration, or an initial seeded administrator user

Source. Own elaboration.

HU-IAM-002: User registration with role assignment

Story

As an administrator or authorized institutional user

I want to register a new user and assign a role

So that the user can exist in the system and access the modules corresponding to their institutional function.

Acceptance criteria

Format: “Given [context/initial state], when [user action], then [expected and verifiable result].”

- [ ] *AC1:* Given that an administrator provides valid user data, when the user registration is submitted, then the system validates that the user does not already exist, creates the user, assigns the selected role, stores the data securely, and shows a confirmation message.
- [ ] *AC2:* Given that the user data is incomplete or invalid, when the registration is submitted, then the system rejects the operation and shows specific validation errors

for the missing or incorrect fields.

- [ ] *AC3:* Given that a user already exists with the same institutional email, identifier, or username, when the administrator attempts to register a new user, then the system prevents duplicate registration and shows a clear error message.
- [ ] *AC4:* Given that a role is selected during registration, when the user is created, then the system associates the user identifier with the person identifier and the assigned role, according to RF 1.1.1.
- [ ] *AC5:* Given that the selected role does not exist or is not allowed for the new user, when the registration is submitted, then the system rejects the operation and shows a validation error.
- [ ] *AC6:* Given that the registration is successful, when the administrator consults the audit or history, then the system records the events `UserRegistered` and `RoleAssigned`.
- [ ] *AC7:* Given normal network conditions, when the administrator completes user registration, then the system responds in less than 5 seconds, according to RNF 1.

Technical notes

- This story covers basic user registration and role assignment. Massive user import by CSV is covered separately in HU-IAM-004.
- The system must validate that the user does not exist before registration, according to RF 1.1 acceptance criteria.
- The system must validate the entered data, according to RF 1.1 acceptance criteria.
- Role assignment must include the relationship between user identifier and person identifier, according to RF 1.1.1.
- User roles considered in the SRS include:
    - Supervisor.
    - Administrator.
    - Teacher / Instructor.

    - Student.
    - Parent or guardian, if applicable.
- *Effective role catalog in the database:* `Administrador | Instructor | Aprendiz` (changeset `auth-004-unify-mobile-roles`); the legacy `SUPER_ADMIN / SCHOOL_ADMIN / INSTRUCTOR / STUDENT` names are merged into those three and are still accepted by the Web client (`backendRoles.js`) for unmigrated databases.
- Password handling must follow security requirements. If registration generates an initial password or invitation flow, this must be defined by the team.
- Sensitive data must be stored securely and must not appear in logs.
- Registration should be auditable for traceability.

Responsible service(s): `ms-identity` (person + user) and `ms-authorization` (role assignment).

Endpoint(s) implemented:

- `POST /api/v1/persons` — implemented (`PersonController`).
- `POST /api/v1/users` — implemented (`UserController`).
- `POST /api/v1/users/{userId}/roles` — implemented in `ms-authorization` (`UserRoleController`), called after user creation.
- `POST /api/v1/users/{id}/activate` · `PATCH /api/v1/users/{id}/status` — implemented (`UserController`).

Events generated:

- `UserRegistered`
- `RoleAssigned`
- `UserRegistrationFailed`

Required permissions:

- Administrator or Supervisor, depending on institutional policy.
- Self-registration should only be allowed if explicitly approved by the institution.

Definition of Done (DoD)

This HU can only be closed when it meets the team's full DoD.

Additional checks specific to this HU:

- [ ] Successful user registration tested.
- [ ] Invalid or incomplete data scenario tested.
- [ ] Duplicate user scenario tested.
- [ ] Role assignment tested.
- [ ] Invalid role scenario tested.
- [ ] User-person-role relationship is correctly stored.
- [ ] Sensitive data is not exposed in logs or UI.
- [ ] Audit events are generated.
- [ ] Confirmation and error messages are clear.
- [ ] Response time is validated under normal conditions and remains under 5 seconds.
- [ ] Traceability to RF 1.1 and RF 1.1.1 is documented.

Estimation and priority

Table 3

Estimation and priority

Field	Value
Story Points	8
Priority	High
Target sprint	Sprint 1 / MVP
Dependencies	Initial role catalog; HU-IAM-003 recommended for permission matrix

Source. Own elaboration.

HU-IAM-003: Role-based permission assignment

Story

As an administrator

I want to assign permissions to each role

So that users can only access the modules, views, and actions allowed by their active role.

Acceptance criteria

Format: “Given [context/initial state], when [user action], then [expected and verifiable result].”

- [ ] *AC1:* Given that an administrator is authenticated, when the administrator assigns permissions to a role, then the system stores the role-permission relationship and applies it to users assigned to that role.
- [ ] *AC2:* Given that a user logs in with an assigned role, when the system loads the user interface or API permissions, then the user only sees and can access the modules and actions allowed for that role.
- [ ] *AC3:* Given that a user attempts to access a module, view, or action not allowed by their role, when the request is processed, then the system denies access and records the unauthorized attempt.
- [ ] *AC4:* Given that an administrator updates the permissions of a role, when the changes are saved, then the new permissions are applied to subsequent permission evaluations (`GET /api/v1/auth/evaluate`) and an audit event is generated.
- [ ] *AC5:* Given that the administrator tries to assign an invalid module, view, or action, when the permission assignment is submitted, then the system rejects the operation and shows a validation error.
- [ ] *AC6:* Given that permissions have been assigned to roles, when an administrator consults the permission matrix, then the system displays the permissions associated with each role.
- [ ] *AC7:* Given normal network conditions, when the administrator consults or updates role permissions, then the system responds in less than 5 seconds, according to RNF 1.

Technical notes

- This story implements the foundation for Role-Based Access Control, according to RF 1.1.2.
- Permission validation must be enforced on the server side. Client-side hiding of options is not sufficient.
- The system should support a role-permission matrix or equivalent authorization model.
- Roles considered in the SRS include:
    - Supervisor.
    - Administrator.
    - Teacher / Instructor.
    - Student.
    - Parent or guardian, if applicable.

  Effective catalog in the database: `Administrador | Instructor | Aprendiz` (`auth-004-unify-mobile-roles`); roles are *global* — `authorization.role` has no `school_id`.
- The permission model should consider modules, views, and actions if the system requires fine-grained access control.
- Unauthorized access attempts should be logged for auditing purposes.
- Permission changes must be traceable and should appear in user-change history or audit logs.
- The system must avoid privilege escalation, meaning users should not be able to grant themselves permissions.

Responsible service(s): `ms-authorization`.

Endpoint(s) implemented:

- `GET /api/v1/roles` · `POST /api/v1/roles` · `GET/PUT/DELETE /api/v1/roles/{roleId}` — implemented (`RoleController`).

- `GET /api/v1/roles/{roleId}/permissions` · `POST /api/v1/roles/{roleId}/permissions` · `DELETE /api/v1/roles/{roleId}/permissions/{permissionId}` — implemented.
- `GET /api/v1/permissions` — implemented (`PermissionController`).
- `GET /api/v1/users/{userId}/roles` · `GET /api/v1/auth/evaluate?userId=&permission=` — implemented (`UserRoleController`); there is *no* `/users/me/permissions` endpoint (clients derive UI permissions from `/users/{id}/roles`).

Events generated:

- `RolePermissionsUpdated`
- `UnauthorizedAccessAttempt`

Required permissions:

- Administrator or Supervisor for role-permission management.
- Regular users may only read their own effective permissions.

Definition of Done (DoD)

This HU can only be closed when it meets the team's full DoD.

Additional checks specific to this HU:

- [ ] Permission assignment by role tested.
- [ ] Permission update tested.
- [ ] Unauthorized access attempt tested.
- [ ] Permission matrix consultation tested.
- [ ] Server-side authorization validation tested.
- [ ] Audit event for permission changes tested.
- [ ] Invalid permission assignment scenario tested.
- [ ] Sensitive authorization data is not exposed in logs.
- [ ] Response time is validated under normal conditions and remains under 5 seconds.
- [ ] Traceability to RF 1.1.2 is documented.

Estimation and priority

Table 4

Estimation and priority

Field	Value
Story Points	5
Priority	High
Target sprint	Sprint 1 / MVP
Dependencies	HU-IAM-002: User registration with role assignment

Source. Own elaboration.

HU-ACAD

HU-ACAD-001: Register environments/classrooms

Story

As an administrator or supervisor

I want to register environments or classrooms

So that attendance records can be associated with the correct training space where students attend class.

Acceptance criteria

Format: “Given [context/initial state], when [user action], then [expected and verifiable result].”

- [ ] *AC1:* Given that an administrator or supervisor is authenticated in the system, when the user submits valid environment identifiers, then the system validates the data, registers the environment/classroom, stores the information securely, and shows a confirmation message.
- [ ] *AC2:* Given that the environment data is incomplete or invalid, when the registration is submitted, then the system rejects the operation and shows specific validation messages.
- [ ] *AC3:* Given that an environment with the same unique identifier already exists, when the administrator attempts to register it again, then the system prevents duplicate registration and shows a clear error message.
- [ ] *AC4:* Given that an environment is successfully registered, when the system requires an environment for attendance registration or reporting, then the registered environment is available for selection.
- [ ] *AC5:* Given normal network conditions, when the administrator registers an environment, then the system responds in less than 5 seconds, according to RNF 1.
- [ ] *AC6:* Given that an environment is registered, when the operation is completed, then the system records an audit event containing the action, user, date, and time.

Technical notes

- The SRS refers to environments as “ambientes/salones”, which are the physical or virtual spaces where attendance is recorded.
- Environment data must be validated before persistence, according to RF 2.1 acceptance criteria.
- The information must be stored securely, according to RF 2.1 and RNF 4.
- This story does not include assignment of schedules, cohorts, or instructors. Those relationships are handled in other ACAD stories.
- The environment entity will be used later by attendance registration, reports, and dashboards.
- The user interface should allow consulting registered environments after creation.
- Auditability is recommended for administrative changes.

Responsible service(s): Academic Structure service, proposed.

Endpoint(s) implemented:

- `POST /api/v1/environments` — proposed.
- `GET /api/v1/environments` — proposed.

Events generated:

- `EnvironmentRegistered`
- `EnvironmentRegistrationFailed`

Required permissions:

- Administrator or Supervisor.

Definition of Done (DoD)

This HU can only be closed when it meets the team's full DoD.

Additional checks specific to this HU:

- [ ] Successful environment registration tested.
- [ ] Invalid or incomplete environment data tested.
- [ ] Duplicate environment scenario tested.

- [ ] Registered environment is available for selection in academic flows.
- [ ] Confirmation and error messages are clear.
- [ ] Audit event is generated.
- [ ] Response time is validated under normal conditions and remains under 5 seconds.
- [ ] Data is stored securely.
- [ ] Traceability to RF 2.1 is documented.

Estimation and priority

Table 5

Estimation and priority

Field	Value
Story Points	3
Priority	High
Target sprint	Sprint 1 / MVP
Dependencies	HU-IAM-001: User login; HU-IAM-003: Role-based permission assignment

Source. Own elaboration.

HU-ACAD-002: Register cohorts/courses

Story

As an administrator or supervisor

I want to register cohorts or courses, known as “fichas” in the SENA context

So that student groups can be associated with attendance control, responsible instructors, and training schedules.

Acceptance criteria

Format: “Given [context/initial state], when [user action], then [expected and verifiable result].”

- [ ] *AC1:* Given that an administrator or supervisor is authenticated in the system, when the user submits valid cohort/course identifiers, then the system validates the

data, registers the cohort/course associated with attendance control, stores the information securely, and shows a confirmation message.

- [ ] *AC2:* Given that the cohort/course data is incomplete or invalid, when the registration is submitted, then the system rejects the operation and shows specific validation messages.
- [ ] *AC3:* Given that a cohort/course with the same unique identifier already exists, when the administrator attempts to register it again, then the system prevents duplicate registration and shows a clear error message.
- [ ] *AC4:* Given that a cohort/course is successfully registered, when attendance registration or academic assignment is performed, then the system can display or use the cohort/course associated with the corresponding user or training group, according to RF 2.2.
- [ ] *AC5:* Given that a cohort/course is successfully registered, when the administrator assigns responsible instructors or schedules, then the cohort/course is available for those relationships.
- [ ] *AC6:* Given normal network conditions, when the administrator registers a cohort/course, then the system responds in less than 5 seconds, according to RNF 1.
- [ ] *AC7:* Given that a cohort/course is registered, when the operation is completed, then the system records an audit event containing the action, user, date, and time.

Technical notes

- The SRS uses the terms “fichas/cursos”. The team must define whether a “ficha” is treated as a cohort, group, or course in the final domain model.
- This story covers the registration of the group/cohort, not the assignment of students, instructors, or schedules.
- The registered cohort/course must be available for later assignment of responsible users, according to RF 2.3.1.

- The cohort/course entity will be used by attendance registration, history queries, justifications, and reports.
- Validation rules should include required identifiers and institutional context.
- Sensitive data is not expected in this entity, but administrative changes should be auditable.
- The system should store data securely, according to RF 2.2 and RNF 4.

Responsible service(s): Academic Structure service, proposed.

Endpoint(s) implemented:

- `POST /api/v1/cohorts` — proposed, representing SENA “fichas”.
- `GET /api/v1/cohorts` — proposed.

Events generated:

- `CohortRegistered`
- `CohortRegistrationFailed`

Required permissions:

- Administrator or Supervisor.

Definition of Done (DoD)

This HU can only be closed when it meets the team's full DoD.

Additional checks specific to this HU:

- [ ] Successful cohort/course registration tested.
- [ ] Invalid or incomplete cohort/course data tested.
- [ ] Duplicate cohort/course scenario tested.
- [ ] Registered cohort/course is available for instructor assignment.
- [ ] Registered cohort/course is available for attendance-related flows.
- [ ] Confirmation and error messages are clear.
- [ ] Audit event is generated.
- [ ] Response time is validated under normal conditions and remains under 5 seconds.
- [ ] Data is stored securely.

- [ ] Traceability to RF 2.2 is documented.

Estimation and priority

Table 6

Estimation and priority

Field	Value
Story Points	3
Priority	High
Target sprint	Sprint 1 / MVP
Dependencies	HU-IAM-001: User login; HU-IAM-003: Role-based permission assignment; HU-ACAD-001 recommended for environment context

Source. Own elaboration.

HU-ACAD-003: Assign responsible instructors

Story

As an administrator or supervisor

I want to assign a teacher/instructor as the responsible user for a cohort or training block

So that attendance records are linked to the correct instructor and instructors can view the groups assigned to them.

Acceptance criteria

Format: “Given [context/initial state], when [user action], then [expected and verifiable result].”

- [ ] *AC1:* Given that an administrator or supervisor has logged into FaceAttend EDU and users are registered in the system, when the user accesses the responsible assignment function, then the system shows the list of users associated with the administrator and the list of users of type teacher/instructor, according to RF 2.3.
- [ ] *AC2:* Given that a valid teacher/instructor and a registered cohort/course are selected, when the administrator assigns the responsible user, then the system

associates the instructor with the cohort/course or training block and shows a confirmation message.

- [ ] *AC3:* Given that the selected user is not a valid user or is not of type teacher/instructor, when the administrator attempts the assignment, then the system rejects the operation and shows a validation error.
- [ ] *AC4:* Given that the cohort/course does not exist in the system, when the administrator attempts to assign a responsible user, then the system rejects the operation and shows a validation error.
- [ ] *AC5:* Given that a responsible instructor is successfully assigned, when attendance is registered for the corresponding cohort or training block, then the system can identify the instructor responsible for that attendance record.
- [ ] *AC6:* Given normal network conditions, when the administrator assigns a responsible instructor, then the system responds in less than 5 seconds, according to RNF 1.
- [ ] *AC7:* Given that a responsible instructor is assigned or updated, when the operation is completed, then the system records an audit event containing the action, administrator, instructor, cohort/course, date, and time.

Technical notes

- The SRS describes this requirement as obtaining the teacher/instructor responsible for the training block corresponding to attendance registration.
- The assignment must require prior login and prior registration in FaceAttend EDU, according to RF 2.3 acceptance criteria.
- The system must show both:
    - Users associated with the administrator.
    - Users of type teacher/instructor.
- This story does not yet include assignment of time blocks or schedules. That is covered in HU-ACAD-005.

- The responsible instructor relationship will be used by attendance, history, reports, and dashboards.
- The system should prevent assigning responsibility to users without an instructor/teacher role.
- Auditability is recommended because this assignment affects academic visibility and reporting permissions.
- The team should define whether a cohort/course can have one or multiple responsible instructors.

Responsible service(s): Academic Structure service, proposed.

Endpoint(s) implemented:

- `GET /api/v1/users?role=instructor` — proposed.
- `POST /api/v1/cohorts/{cohortId}/responsible` — proposed.
- `GET /api/v1/cohorts/{cohortId}/responsible` — proposed.

Events generated:

- `ResponsibleAssigned`
- `ResponsibleAssignmentFailed`

Required permissions:

- Administrator or Supervisor.

Definition of Done (DoD)

This HU can only be closed when it meets the team's full DoD.

Additional checks specific to this HU:

- [ ] Administrator login tested before assignment.
- [ ] Instructor user list display tested.
- [ ] Users associated with the administrator display tested.
- [ ] Successful responsible instructor assignment tested.
- [ ] Invalid instructor scenario tested.
- [ ] Non-existent cohort/course scenario tested.

- [ ] Assigned instructor is correctly linked to the cohort/course or training block.
- [ ] Audit event is generated.
- [ ] Confirmation and error messages are clear.
- [ ] Response time is validated under normal conditions and remains under 5 seconds.
- [ ] Traceability to RF 2.3 is documented.

Estimation and priority

Table 7

Estimation and priority

Field	Value
Story Points	5
Priority	High
Target sprint	Sprint 1 / MVP
Dependencies	HU-IAM-001: User login; HU-IAM-002: User registration with role assignment; HU-ACAD-002: Register cohorts/courses

Source. Own elaboration.

HU-ACAD-004: Assign cohorts to responsible users

Story

As an administrator or supervisor

I want to assign registered cohorts/courses, known as “fichas”, to a valid responsible user

So that administrative users, instructors, and responsible users can view and manage the cohorts assigned to their academic responsibility.

Acceptance criteria

Format: “Given [context/initial state], when [user action], then [expected and verifiable result].”

- [ ] *AC1:* Given that an administrator or supervisor is authenticated in the system, when the user selects a valid responsible user and an existing cohort/course, then the

system associates the cohort/course with the responsible user and shows a confirmation message.

- [ ] *AC2:* Given that the selected responsible user is not a valid registered user, when the assignment is submitted, then the system rejects the operation and shows a validation error.
- [ ] *AC3:* Given that the selected cohort/course does not exist in the system, when the assignment is submitted, then the system rejects the operation and shows a validation error.
- [ ] *AC4:* Given that a responsible user has assigned cohorts/courses, when the user requests the list of cohorts under their responsibility, then the system displays the cohorts corresponding to their assigned responsibility, according to RF 2.3.1.
- [ ] *AC5:* Given that an administrative user, supervisor, administrator, or teacher/instructor accesses the cohort assignment module, when the system loads the information, then the user sees only the cohorts allowed by their role.
- [ ] *AC6:* Given normal network conditions, when the administrator assigns or consults cohorts for a responsible user, then the system responds in less than 5 seconds, according to RNF 1.
- [ ] *AC7:* Given that a cohort/course is assigned to a responsible user, when the operation is completed, then the system records an audit event containing the action, user, responsible user, cohort/course, date, and time.

Technical notes

- The SRS indicates that administrative users (Supervisor, Administrator, and Teacher/Instructor) must be able to obtain the list of cohorts corresponding to their work.
- The system must validate that the responsible user is valid and that the cohort/course exists, according to RF 2.3.1 acceptance criteria.

- This story complements HU-ACAD-003 by formalizing the visibility and association of cohorts assigned to a responsible user.
- The responsible-cohort relationship will be used by attendance registration, history, reports, dashboards, and instructor permissions.
- The system should define whether a responsible user can have one or multiple cohorts assigned.
- The system should prevent assigning inactive users as responsible users.
- Role-based access must be enforced on the server side.
- Auditability is recommended because this assignment affects academic visibility and reporting permissions.

Responsible service(s): Academic Structure service, proposed.

Endpoint(s) implemented:

- `POST /api/v1/responsibles/{responsibleId}/cohorts` — proposed.
- `GET /api/v1/responsibles/{responsibleId}/cohorts` — proposed.
- `GET /api/v1/users/me/cohorts` — proposed.

Events generated:

- `CohortAssignedToResponsible`
- `CohortAssignmentFailed`

Required permissions:

- Administrator or Supervisor for assigning cohorts to responsible users.
- Teacher/Instructor may consult their own assigned cohorts.

Definition of Done (DoD)

This HU can only be closed when it meets the team's full DoD.

Additional checks specific to this HU:

- [ ] Successful cohort assignment to a responsible user tested.
- [ ] Invalid responsible user scenario tested.
- [ ] Non-existent cohort/course scenario tested.

- [ ] Consultation of assigned cohorts tested.
- [ ] Role-based visibility tested for administrator, supervisor, and instructor.
- [ ] Audit event is generated.
- [ ] Confirmation and error messages are clear.
- [ ] Response time is validated under normal conditions and remains under 5 seconds.
- [ ] Traceability to RF 2.3.1 is documented.

Estimation and priority

Table 8

Estimation and priority

Field	Value
Story Points	5
Priority	High
Target sprint	Sprint 1 / MVP
Dependencies	HU-IAM-001: User login; HU-ACAD-002: Register cohorts/courses; HU-ACAD-003: Assign responsible instructors

Source. Own elaboration.

conference

HU-ACAD-005: Assign conference/time blocks

Story

As an administrator or supervisor

I want to assign or time blocks to a registered cohort with a valid responsible instructor

So that the system knows the schedule in which students must attend and register their attendance.

Acceptance criteria

Format: “Given [context/initial state], when [user action], then [expected and verifiable result].”

- [ ] *AC1:* Given that an administrator or supervisor is authenticated in the system, when the user assigns a time block to a registered cohort with a valid responsible instructor, then the system associates the time block with the cohort and responsible user and shows a confirmation message.
- [ ] *AC2:* Given that the selected responsible user is not valid or is not registered, when the time block assignment is submitted, then the system rejects the operation and shows a validation error.
- [ ] *AC3:* Given that the proposed time block overlaps with another time block already assigned, when the assignment is submitted, then the system rejects the operation and shows a clear conflict message.
- [ ] *AC4:* Given that the cohort is not available for the proposed time block, when the assignment is submitted, then the system rejects the operation and shows a validation error.
- [ ] *AC5:* Given that a jornada/time block is successfully assigned, when attendance registration occurs for that cohort, then the system can identify the expected time in which students must register entry and exit.
- [ ] *AC6:* Given that users need to consult schedules, when an administrator, instructor, or student requests the corresponding schedule, then the system displays the schedules where users must attend and register, according to their role.
- [ ] *AC7:* Given normal network conditions, when the administrator assigns or consults conference/time blocks, then the system responds in less than 5 seconds, according to RNF 1.
- [ ] *AC8:* Given that a jornada/time block is assigned, updated, or rejected, when the operation is completed, then the system records an audit event containing the action, user, cohort, responsible user, time block, date, and time.

Technical notes

- The SRS refers to “asignar conference” and “tiempo bloque de laburo”. For consistency, this story uses “jornada” or “time block” as the period during which a cohort must attend with a responsible instructor.
- The system must validate that:
    - The responsible user is valid.
    - The time block does not overlap with another already assigned block.
    - The cohort has availability for the assigned time block.
- This story is essential for attendance validation because HU-ATT-001 depends on knowing whether entry and exit times are appropriate for the assigned block.
- The schedule/jornada relationship will be used by attendance registration, late detection, absence detection, history, reports, and dashboards.
- The system should prevent duplicate or conflicting schedules for the same cohort, responsible user, environment, or time period.
- The team must define whether overlap validation applies only to the same cohort, to the same responsible instructor, to the same environment, or to a combination of these.
- Role-based access must be enforced:
    - Administrators and supervisors can assign and consult schedules.
    - Instructors can consult schedules assigned to them.
    - Students can consult schedules where they are registered, if applicable.

Responsible service(s): Academic Structure service / Scheduling service, proposed.

Endpoint(s) implemented:

- `POST /api/v1/schedules` — proposed.
- `GET /api/v1/schedules` — proposed.
- `GET /api/v1/cohorts/{cohortId}/schedule` — proposed.
- `GET /api/v1/users/me/schedule` — proposed.

Events generated:

- `ScheduleAssigned`
- `ScheduleAssignmentFailed`
- `ScheduleConflictDetected`

Required permissions:

- Administrator or Supervisor for assigning conference/time blocks.
- Teacher/Instructor may consult schedules assigned to them.
- Student may consult allowed schedules if applicable.

Definition of Done (DoD)

This HU can only be closed when it meets the team's full DoD.

Additional checks specific to this HU:

- [ ] Successful jornada/time block assignment tested.
- [ ] Invalid responsible user scenario tested.
- [ ] Time block overlap scenario tested.
- [ ] Cohort unavailable scenario tested.
- [ ] Assigned time block is correctly associated with cohort and responsible user.
- [ ] Schedule consultation tested for administrator, instructor, and student where applicable.
- [ ] Attendance module can use the assigned time block for entry/exit validation.
- [ ] Audit event is generated.
- [ ] Confirmation and error messages are clear.
- [ ] Response time is validated under normal conditions and remains under 5 seconds.
- [ ] Traceability to RF 2.3.2 is documented.

Estimation and priority

Table 9

Estimation and priority

Field	Value
Story Points	8
Priority	High
Target sprint	Sprint 1 / MVP
Dependencies	HU-IAM-001: User login; HU-ACAD-003: Assign responsible instructors; HU-ACAD-004: Assign cohorts to responsible users

Source. Own elaboration.

HU-ATT

HU-ATT-001: Facial attendance registration

Story

As a student/learner

I want to register my entry using facial recognition

So that my attendance is recorded quickly, securely, and without depending on manual roll call by the instructor.

Acceptance criteria

Format: "Given [context/initial state], when [user action], then [expected and verifiable result]."

- [ ] *AC1:* Given that the student is registered, active, has an active facial parameter, and has an assigned training block, when the student scans their face during the allowed entry period, then the system registers the entry, shows a confirmation message, and stores the date, time, training block, and registration method as "facial".
- [ ] *AC2:* Given that the facial scan matches a registered user, when the system evaluates the entry time against the assigned training block, then the attendance record is marked as present or late according to the attendance rules configured in the system.
- [ ] *AC3:* Given that the student already has an entry record for the same training block and date, when the student attempts to register entry again, then the system prevents duplicate registration and shows a clear message indicating that the entry was already registered.
- [ ] *AC4:* Given that the student is not registered, is inactive, or does not have an active facial parameter, when the student attempts facial attendance registration, then the system denies the registration and shows a clear error message.
- [ ] *AC5:* Given that the face is not detected, does not match a registered user, or the camera fails, when the facial scan fails, then the system shows an error message and

offers the corresponding alternative flow according to RF 3.1.2 or RF 3.2.1.

- [ ] *AC6:* Given normal network, camera, and lighting conditions, when the student completes facial attendance registration, then the system responds in less than 5 seconds, according to RNF 1.
- [ ] *AC7:* Given that biometric data is processed, when the registration is completed, then the system records an audit event and does not expose sensitive biometric data in logs, alerts, or the user interface.
- [ ] *AC8:* Given that the student exits the training session, when the student scans their face during the allowed exit period, then the system registers the exit, shows a confirmation message, and stores the exit time without duplicating the entry record.

Technical notes

- Facial recognition may depend on an external cloud service, according to SRS assumption 2.4.5. If the external service fails, the system must show a clear error and allow the corresponding alternative flow.
- The system requires internet access, functional camera, and adequate lighting and face positioning, according to SRS constraints and assumptions.
- The system must validate:
    - The user exists and is active.
    - The user has an active facial parameter.
    - The user has an assigned training block or schedule.
    - The entry or exit time is appropriate for the assigned block.
    - The user does not already have a duplicate entry or exit record for the same block and date.
- Alternative registration for registered users whose facial scan fails is covered by RF 3.1.2 and should be linked from the failure state.
- Unregistered or casual users are covered by RF 3.2 and RF 3.2.1, but they should not be mixed with regular student attendance records without explicit classification.

- Biometric data must be treated as sensitive data and must comply with applicable data-protection regulations, such as Colombian Law 1581 of 2012.
- The system should prefer facial templates or embeddings over raw facial images, unless raw images are explicitly required and approved.
- Sensitive data must not be written to logs or shown in the interface.
- The attendance registration flow must generate audit information for traceability.
- Performance requirement: response time under 5 seconds, according to RNF 1.

Responsible service(s): Attendance service / Facial verification service, proposed.

Endpoint(s) implemented:

- `POST /api/v1/attendance/entry/facial` — proposed.
- `POST /api/v1/attendance/exit/facial` — proposed.
- `POST /api/v1/attendance/entry/alternative` — fallback, may belong to HU-ATT-002.

Events generated:

- `AttendanceEntryRegistered`
- `AttendanceExitRegistered`
- `LateDetected`
- `FacialScanFailed`
- `AlternativeRegistrationRequested`

Required permissions:

- Authenticated student/learner with active account.
- The system must additionally validate active facial parameter and assigned training block.

Definition of Done (DoD)

This HU can only be closed when it meets the team's full DoD.

See: `00-governance/definition-of-done.md` (`fae-docs` repository).

Additional checks specific to this HU:

- [ ] Facial scan success flow tested with a registered and active user.
- [ ] Facial scan failure flow tested when the face is not detected.
- [ ] Facial scan failure flow tested when the face does not match a registered user.
- [ ] Camera unavailable or permission denied scenario tested.
- [ ] Duplicate entry prevention tested.
- [ ] Duplicate exit prevention tested.
- [ ] Late detection tested when entry occurs after the allowed time.
- [ ] Alternative registration flow is reachable when facial scan fails.
- [ ] Response time is validated under normal conditions and remains under 5 seconds.
- [ ] Sensitive biometric data is not exposed in logs, responses, or UI.
- [ ] Audit event is generated for successful and failed registration attempts.
- [ ] Confirmation and error messages are clear, visible, and understandable.
- [ ] Traceability to RF 3.1, RF 3.1.1, CU 1, and CU 2 is documented.

Estimation and priority

Table 10

Estimation and priority

Field	Value
Story Points	8
Priority	High
Target sprint	Sprint 1 / MVP
Dependencies	HU-IAM-001: User login; HU-IAM-002: User registration with role assignment; HU-ATT-002: Initial facial parameter enrollment; HU-ACAD-005: Assign conference/time blocks

Source. Own elaboration.

HU-ATT-002: Initial facial parameter enrollment

Story

As a registered student/learner

I want to register my facial parameter in FaceAttend EDU

So that I can later register my attendance using facial recognition without depending on manual roll call.

Acceptance criteria

Format: “Given [context/initial state], when [user action], then [expected and verifiable result].”

- [ ] *AC1:* Given that the user is registered and has started a session in FaceAttend EDU, when the user initiates the facial enrollment process, then the system activates the camera and prepares for face capture, according to Use Case 4.
- [ ] *AC2:* Given that the camera is active and the user positions their face correctly, when the system captures the facial image under adequate lighting and positioning conditions, then the system processes the capture and associates the resulting facial parameter with the authenticated user.
- [ ] *AC3:* Given that the facial capture is successful, when the enrollment process finishes, then the system stores the facial parameter securely, shows a confirmation message, and records an audit event.
- [ ] *AC4:* Given that the face is not detected, the image quality is insufficient, or the camera fails, when the enrollment attempt fails, then the system shows a clear error message and allows the user to retry.
- [ ] *AC5:* Given that the user already has an active facial parameter, when the user attempts initial enrollment again, then the system informs that a facial parameter already exists and redirects the user to the update flow, unless replacement is explicitly allowed and confirmed.
- [ ] *AC6:* Given that biometric data is processed, when the facial parameter is stored, then the system protects sensitive data according to RNF 4 and does not expose raw biometric information in logs or user interface messages.
- [ ] *AC7:* Given normal device, camera, and network conditions, when the user completes facial enrollment, then the system responds in less than 5 seconds,

according to RNF 1.

Technical notes

- This story corresponds to the initial facial enrollment required before using facial attendance registration in HU-ATT-001.
- The SRS indicates that facial recognition may fail due to lighting or user movement, so the interface must provide clear guidance.
- Facial enrollment may depend on an external cloud facial recognition service, according to SRS assumption 2.4.5. If the external service fails, the system must show a clear error and allow retry.
- The system should store facial templates or parameters instead of raw facial images unless raw images are explicitly required and approved.
- Biometric data must comply with applicable data-protection regulations, such as Colombian Law 1581 of 2012.
- The user must be authenticated before associating a facial parameter with their account.
- Camera permissions must be requested and handled gracefully if denied.
- Facial parameter updates are covered separately in HU-CONF-006.
- Sensitive biometric data must not be written to logs or shown in the interface.

Responsible service(s): Attendance service / Biometric enrollment service, proposed.

Endpoint(s) implemented:

- `POST /api/v1/biometrics/facial-enrollment` — proposed.
- `GET /api/v1/users/me/facial-status` — proposed.

Events generated:

- `FacialParameterRegistered`
- `FacialRegistrationFailed`
- `FacialParameterAlreadyExists`

Required permissions:

- Authenticated student/learner.
- Administrator or Supervisor may assist with enrollment only if allowed by institutional policy.

Definition of Done (DoD)

This HU can only be closed when it meets the team's full DoD.

Additional checks specific to this HU:

- [ ] Successful facial enrollment tested.
- [ ] Face not detected scenario tested.
- [ ] Poor lighting or movement scenario tested.
- [ ] Camera unavailable or permission denied scenario tested.
- [ ] Existing facial parameter scenario tested.
- [ ] Facial parameter is correctly associated with the authenticated user.
- [ ] Sensitive biometric data is not exposed in logs, responses, or UI.
- [ ] Audit event is generated.
- [ ] Confirmation and error messages are clear.
- [ ] Response time is validated under normal conditions and remains under 5 seconds.
- [ ] Traceability to Use Case 4 and related RFs is documented.

Estimation and priority

Table 11

Estimation and priority

Field	Value
Story Points	8
Priority	High
Target sprint	Sprint 1 / MVP
Dependencies	HU-IAM-001: User login; HU-IAM-002: User registration with role assignment

Source. Own elaboration.

HU-ATT-003: Alternative registration for registered users with facial scan failure

Story

As a registered student/learner whose facial scan fails

I want to complete a time-limited alternative registration process

So that my attendance can still be recorded even when facial recognition cannot validate me.

Acceptance criteria

Format: “Given [context/initial state], when [user action], then [expected and verifiable result].”

- [ ] *AC1:* Given that a registered user attempted facial attendance registration and the facial scan failed, when the user selects the alternative registration option, then the system presents a time-limited survey requesting the personal data required for validation, according to RF 3.1.2.
- [ ] *AC2:* Given that the user provides personal data within the configured time limit, when the alternative registration is submitted, then the system compares the provided data with the data registered in the system.
- [ ] *AC3:* Given that the provided data matches the registered data, when the comparison is completed, then the system registers attendance, shows a confirmation message, stores the record with the method marked as “alternative”, and records an audit event.
- [ ] *AC4:* Given that the provided data does not match the registered data, when the comparison is completed, then the system rejects the alternative registration and shows a clear error message.
- [ ] *AC5:* Given that the configured time limit expires before the user submits the survey, when the user attempts to continue, then the system cancels the alternative registration attempt and shows a timeout message.
- [ ] *AC6:* Given that the person attempting alternative registration is not a registered user, when the alternative registration data is validated, then the system rejects the

process and directs the person to the unregistered-user flow, if applicable.

- [ ] *AC7:* Given normal system conditions, when the alternative registration request is processed, then the system responds in less than 5 seconds, according to RNF 1.

Technical notes

- The SRS describes the alternative registration as a survey with a time limit, where the user enters personal data to be compared with registered data.
- The time limit value must be defined by the team. A proposed initial value is 60 seconds, but it must be validated.
- The personal data fields used for validation must be defined by the team and should follow data-minimization principles.
- This story is a fallback for HU-ATT-001 when facial recognition fails due to lighting, movement, camera failure, or non-matching facial parameters.
- The system must validate that the user is registered in the system, according to RF 3.1.2.
- The alternative registration must be linked to the corresponding attendance block, date, and user.
- The system should prevent abuse of the alternative registration flow using rate limiting or attempt controls.
- Personal data used for validation must not be written to logs or shown in the interface.
- If the alternative flow is used in a kiosk or shared device, the system must not expose unrelated user information.
- This story does not cover unregistered or NN users. That case is handled by HU-ATT-004 and HU-ATT-005.

Responsible service(s): Attendance service / Alternative validation service, proposed.

Endpoint(s) implemented:

- `POST /api/v1/attendance/alternative/validate` — proposed.
- `POST /api/v1/attendance/alternative/register` — proposed.

Events generated:

- `AlternativeAttendanceRegistered`
- `AlternativeRegistrationFailed`
- `AlternativeRegistrationTimedOut`
- `UnregisteredUserDetected`

Required permissions:

- Registered user whose data can be validated.
- If used in a kiosk or attendance terminal, the endpoint should be restricted to the attendance context and protected with rate limiting.

Definition of Done (DoD)

This HU can only be closed when it meets the team's full DoD.

Additional checks specific to this HU:

- [ ] Alternative registration flow is reachable after facial scan failure.
- [ ] Successful alternative registration tested.
- [ ] Data mismatch scenario tested.
- [ ] Time limit expiration scenario tested.
- [ ] Unregistered user scenario tested.
- [ ] Attendance record is stored with method marked as alternative.
- [ ] Personal data used for validation is not exposed in logs or UI.
- [ ] Audit event is generated.
- [ ] Confirmation and error messages are clear.
- [ ] Response time is validated under normal conditions and remains under 5 seconds.
- [ ] Traceability to RF 3.1.2 is documented.

Estimation and priority

Table 12

Estimation and priority

Field	Value
Story Points	5
Priority	High
Target sprint	Sprint 1 / MVP
Dependencies	HU-IAM-002: User registration with role assignment; HU-ATT-001: Facial attendance registration; HU-ACAD-005: Assign conference/time blocks

Source. Own elaboration.

HU-ATT-004: Entry control for unregistered or NN users

Story

As an administrator, supervisor, or attendance-control user

I want the system to classify people who are not part of the official attendance control as NN or casual users

So that the system can distinguish official registered users from visitors or unregistered people.

Acceptance criteria

Format: “Given [context/initial state], when [user action], then [expected and verifiable result].”

- [ ] *AC1:* Given that a person attempts to be registered in the attendance control process, when the system checks the person’s identification data, then the system determines whether the person is a registered user or an unregistered/NN user.
- [ ] *AC2:* Given that the person is registered in FaceAttend EDU, when the classification process is executed, then the system assigns the person the corresponding registered-user type and directs the process to the normal or alternative registered-user flow.
- [ ] *AC3:* Given that the person is not registered in FaceAttend EDU, when the classification process is executed, then the system assigns the person the NN or casual-user type and enables the corresponding unregistered-user flow.

- [ ] *AC4:* Given that the identification data is incomplete, invalid, or cannot be processed, when the classification request is submitted, then the system rejects the classification and shows a clear validation message.
- [ ] *AC5:* Given that a person is classified as NN or casual user, when the classification is completed, then the system stores the classification separately from official student attendance records and records an audit event.
- [ ] *AC6:* Given normal system conditions, when the classification request is processed, then the system responds in less than 5 seconds, according to RNF 1.

Technical notes

- The SRS RF 3.2 describes this requirement as determining which users are target users of attendance control and which are casual users.
- The output defined by the SRS is the assignment of user type: NN/user.
- The SRS acceptance criterion says the system must validate whether the user is registered. For this story, that validation is interpreted as checking whether the person exists in the official user registry.
- This story covers classification and control. The actual alternative registration for NN users is handled in HU-ATT-005.
- The system must avoid mixing official attendance records with casual or NN records unless the institution explicitly requires consolidated reporting.
- Personal data used to classify a person must be minimized and protected.
- The classification process may be used in kiosk, reception, or attendance-terminal scenarios.
- Rate limiting and auditability are recommended to prevent abuse or unauthorized queries.
- The team must define which identification fields are required to classify a person.

Responsible service(s): Attendance service / Person classification service, proposed.

Endpoint(s) implemented:

- `POST /api/v1/attendance/person-classification` — proposed.
- `GET /api/v1/attendance/classifications` — proposed for administrative review.

Events generated:

- `PersonClassifiedAsRegisteredUser`
- `PersonClassifiedAsNN`
- `PersonClassificationFailed`

Required permissions:

- System process for automatic classification.
- Administrator or Supervisor for reviewing classifications.
- Kiosk or attendance-terminal access should be restricted and rate-limited.

Definition of Done (DoD)

This HU can only be closed when it meets the team's full DoD.

Additional checks specific to this HU:

- [ ] Registered user classification tested.
- [ ] Unregistered/NN user classification tested.
- [ ] Invalid or incomplete identification data tested.
- [ ] Classification is stored separately from official attendance records.
- [ ] NN classification enables the corresponding unregistered-user flow.
- [ ] Registered user classification redirects to normal or alternative registered flow.
- [ ] Audit event is generated.
- [ ] Personal data is not exposed in logs or UI.
- [ ] Confirmation and error messages are clear.
- [ ] Response time is validated under normal conditions and remains under 5 seconds.
- [ ] Traceability to RF 3.2 is documented.

Estimation and priority

Table 13

Estimation and priority

Field	Value
Story Points	5
Priority	High
Target sprint	Sprint 2
Dependencies	HU-IAM-002: User registration with role assignment; HU-ATT-001: Facial attendance registration; related to HU-ATT-005

Source. Own elaboration.

HU-JUS

HU-JUS-001: Justification support upload

Story

As a student/learner

I want to upload a justification with its corresponding support document for an absence or late arrival

So that my case can be formally reviewed and evaluated by a superior, instead of remaining as an unjustified absence or late arrival.

Acceptance criteria

Format: "Given [context/initial state], when [user action], then [expected and verifiable result]."

- [ ] *AC1:* Given that the student is authenticated and has at least one recorded absence or late arrival pending justification, when the student accesses the justification module, then the system displays the list of absences or late arrivals eligible for justification.
- [ ] *AC2:* Given that the student selects an eligible absence or late arrival, when the student fills in the justification reason and uploads a valid support document, then the system validates the data, stores the justification with status "Pending", associates it with the corresponding attendance record, and shows a confirmation message.
- [ ] *AC3:* Given that the student attempts to submit a justification without uploading a support document, when the submission is processed, then the system rejects the operation and shows a clear validation message indicating that a support document is required.
- [ ] *AC4:* Given that the uploaded support document exceeds the maximum allowed file size or has an unsupported format, when the upload is attempted, then the system rejects the file and shows a clear error message with the allowed formats and maximum size.

- [ ] *AC5:* Given that a justification already exists for the selected absence or late arrival, when the student attempts to submit a new justification for the same record, then the system prevents duplicate justification and shows a clear message indicating that the record already has a justification in progress or resolved.
- [ ] *AC6:* Given that the student does not have any recorded absences or late arrivals eligible for justification, when the student accesses the justification module, then the system shows a clear message indicating that there are no records pending justification.
- [ ] *AC7:* Given that the justification is successfully submitted, when the operation is completed, then the system records an audit event containing the student identifier, the attendance record associated, the justification identifier, the support document reference, the submission date and time, and the initial status as "Pending".
- [ ] *AC8:* Given normal network conditions, when the student submits a justification with its support document, then the system responds in less than 10 seconds, including file upload time.
- [ ] *AC9:* Given that the justification is successfully submitted, when the system processes the submission, then the support document is stored securely and its content is not exposed in logs or user interface messages beyond the file name or reference.

Technical notes

- The SRS Use Case 3 describes the justification flow as follows:
    1. The user registers attendance.
    2. The system informs the user of recorded absences during the established period.
    3. The system asks the user to upload a valid justification for the absence.
    4. If the user does not upload a justification, the absence remains marked as unjustified.

- RF 6.1 indicates that the support document must be uploaded successfully before continuing with the evaluation process.
- The system must validate:
    - The student is authenticated and active.
    - The attendance record exists and corresponds to an absence or late arrival.
    - The attendance record does not already have an active or resolved justification.
    - The support document is present, within the allowed file size, and in a supported format.
    - The justification reason is not empty.
- Supported file formats and maximum file size must be defined by the team. Proposed initial values:
    - Formats: PDF, JPG, JPEG, PNG.
    - Maximum size: 5 MB.
- The justification must be linked to a single attendance record, according to the domain rule RN-32 defined in `entities-and-rules.md`.
- A justification cannot be approved without at least one valid support document, according to domain rule RN-33.
- The support document must be stored securely and must not expose its content in logs or the interface.
- Justification parametrization, such as valid justification types, is managed in HU-CONF-008 and RF 4.7.
- This story does not cover the approval/rejection workflow. That is handled in HU-JUS-002.
- This story does not cover the automatic notification of the result. That is handled in HU-JUS-003.

Responsible service(s): Justification service / File storage service, proposed.

Endpoint(s) implemented:

- `GET /api/v1/attendance/unjustified` — proposed, returns absences and late arrivals eligible for justification.
- `POST /api/v1/justifications` — proposed, creates the justification with the support document.
- `GET /api/v1/justifications/me` — proposed, returns the student's own justifications.

Events generated:

- `JustificationSubmitted`
- `JustificationSubmissionFailed`
- `SupportDocumentUploaded`

Required permissions:

- Authenticated student/learner with at least one unjustified absence or late arrival.

Definition of Done (DoD)

This HU can only be closed when it meets the team's full DoD.

Additional checks specific to this HU:

- [ ] Eligible absences and late arrivals list is displayed correctly.
- [ ] Successful justification submission with valid support document tested.
- [ ] Missing support document scenario tested.
- [ ] Invalid file format scenario tested.
- [ ] Oversized file scenario tested.
- [ ] Duplicate justification prevention tested.
- [ ] No eligible records scenario tested.
- [ ] Justification is correctly linked to the attendance record.
- [ ] Justification is stored with initial status "Pending".
- [ ] Support document is stored securely.

- [ ] Sensitive document content is not exposed in logs or UI.
- [ ] Audit event is generated.
- [ ] Confirmation and error messages are clear.
- [ ] Response time is validated under normal conditions.
- [ ] Traceability to RF 6.1 and Use Case 3 is documented.

Estimation and priority

Table 14

Estimation and priority

Field	Value
Story Points	8
Priority	High
Target sprint	Sprint 2
Dependencies	HU-IAM-001: User login; HU-ATT-001: Facial attendance registration; HU-HIST-001: Attendance and absence query by parameters; HU-CONF-008: Delay and justification parametrization

Source. Own elaboration.

HU-JUS-002: Justification approval and rejection

Story

As an instructor, administrator, or supervisor

I want to review a student's justification, evaluate the uploaded support document, and approve or reject the justification

So that the student's attendance record is updated fairly and transparently according to institutional rules.

Acceptance criteria

Format: "Given [context/initial state], when [user action], then [expected and verifiable result]."

- [ ] *AC1:* Given that an authorized user with a superior role is authenticated, when the user accesses the justification review module, then the system displays the list of

justifications with status "Pending" that correspond to their assigned cohorts, environments, or institutional scope according to their role.

- [ ] *AC2:* Given that the reviewer selects a pending justification, when the system displays the justification detail, then the reviewer can see the student's name, the attendance record date, the absence or late arrival type, the justification reason, and the uploaded support document.
- [ ] *AC3:* Given that the reviewer reviews the support document and determines it is valid according to the configured justification parameters, when the reviewer approves the justification, then the system updates the justification status to "Approved", updates the corresponding attendance record status to "Justified", stores the reviewer's identifier and evaluation date, and shows a confirmation message.
- [ ] *AC4:* Given that the reviewer reviews the support document and determines it is not valid, when the reviewer rejects the justification, then the system updates the justification status to "Rejected", keeps the attendance record in its original status, stores the reviewer's identifier and evaluation date, and shows a confirmation message.
- [ ] *AC5:* Given that the support document does not exist or cannot be retrieved at the time of review, when the reviewer attempts to evaluate the justification, then the system prevents the evaluation and shows a clear error message indicating that the support document is not available.
- [ ] *AC6:* Given that a justification has already been approved or rejected, when another reviewer or the same reviewer attempts to evaluate it again, then the system prevents re-evaluation and shows a clear message indicating that the justification has already been resolved.
- [ ] *AC7:* Given that the user attempting to evaluate the justification does not have a superior role or does not have permission to review justifications for that cohort or

student, when the user attempts the evaluation, then the system denies access and records an unauthorized access attempt.

- [ ] *AC8:* Given that the justification is approved or rejected, when the operation is completed, then the system records an audit event containing the justification identifier, the attendance record, the reviewer identifier, the decision (approved or rejected), and the evaluation date and time.
- [ ] *AC9:* Given normal network conditions, when the reviewer submits the approval or rejection, then the system responds in less than 5 seconds, according to RNF 1.

Technical notes

- RF 6.2 indicates that the support must be validated according to what is defined by the superior.
- RF 4.7 defines the parametrization of valid justification types. The evaluation should consider whether the submitted justification matches the configured valid types.
- Domain rule RN-35 states that a justification can only be evaluated by a user with a role superior to the student's role.
- Domain rule RN-34 states that a justification can only be approved if it meets the criteria parametrized by the administrator.
- Domain rule RN-33 states that a justification cannot be approved if it does not have at least one valid support document.
- The system must validate:
    - The reviewer is authenticated and has a superior role.
    - The reviewer has visibility of the justification according to their assigned cohorts or scope.
    - The justification exists and is in status "Pending".
    - The support document exists and is accessible.
    - The justification has not been previously resolved.

- When a justification is approved, the corresponding attendance record should be updated to "Justified".
- When a justification is rejected, the attendance record should remain in its original status ("Absent" or "Late").
- The reviewer's identity and decision must be stored for traceability.
- The review process must not allow the same justification to be evaluated more than once.
- This story does not cover automatic notification of the result. That is handled in HU-JUS-003.

Responsible service(s): Justification service / Attendance service, proposed.

Endpoint(s) implemented:

- `GET /api/v1/justifications?status=pending` — proposed, returns pending justifications filtered by reviewer scope.
- `GET /api/v1/justifications/{justificationId}` — proposed, returns justification detail with support document.
- `POST /api/v1/justifications/{justificationId}/approve` — proposed.
- `POST /api/v1/justifications/{justificationId}/reject` — proposed.

Events generated:

- `JustificationApproved`
- `JustificationRejected`
- `AttendanceRecordUpdatedToJustified`
- `JustificationEvaluationFailed`
- `UnauthorizedJustificationAccessAttempt`

Required permissions:

- Instructor, Administrator, or Supervisor, depending on institutional policy.
- The reviewer must have visibility of the student's cohort or scope.
- Students must not be able to evaluate their own justifications.

Definition of Done (DoD)

This HU can only be closed when it meets the team's full DoD.

Additional checks specific to this HU:

- [ ] Pending justification list is displayed correctly for the reviewer's scope.
- [ ] Justification detail with support document is displayed correctly.
- [ ] Successful approval tested.
- [ ] Successful rejection tested.
- [ ] Attendance record is updated to "Justified" after approval.
- [ ] Attendance record remains unchanged after rejection.
- [ ] Missing or inaccessible support document scenario tested.
- [ ] Already resolved justification scenario tested.
- [ ] Unauthorized reviewer scenario tested.
- [ ] Student self-evaluation prevention tested.
- [ ] Reviewer identity and decision date are stored.
- [ ] Audit event is generated for approval and rejection.
- [ ] Confirmation and error messages are clear.
- [ ] Response time is validated under normal conditions and remains under 5 seconds.
- [ ] Traceability to RF 6.2, RF 4.7, and Use Case 3 is documented.

Estimation and priority

Table 15

Estimation and priority

Field	Value
Story Points	8
Priority	High
Target sprint	Sprint 2
Dependencies	HU-JUS-001: Justification support upload; HU-CONF-008: Delay and justification parametrization; HU-IAM-003: Role-based permission assignment

Source. Own elaboration.

HU-JUS-003: Automatic notification of justification result

Story

As a student/learner who submitted a justification

I want to receive an automatic notification with the result of my justification (approved or rejected)

So that I know the outcome without having to check the system manually and I can take further action if needed.

Acceptance criteria

Format: "Given [context/initial state], when [user action], then [expected and verifiable result]."

- [ ] *AC1:* Given that a reviewer has approved a justification, when the approval is processed, then the system automatically sends a notification to the student who submitted the justification, including the justification identifier, the attendance record date, and the result as "Approved".
- [ ] *AC2:* Given that a reviewer has rejected a justification, when the rejection is processed, then the system automatically sends a notification to the student who submitted the justification, including the justification identifier, the attendance record date, and the result as "Rejected".
- [ ] *AC3:* Given that the notification is sent, when the student accesses the notification module or their notification inbox, then the student can see the notification with the justification result, the date of the evaluation, and the associated attendance record.
- [ ] *AC4:* Given that the notification fails to be delivered due to a system error, when the delivery fails, then the system records the failure in the platform log and retries delivery according to the configured retry policy, or marks the notification as failed for administrative review.
- [ ] *AC5:* Given that a justification is evaluated, when the result notification is generated, then the notification does not include sensitive details such as the support

document content, the reviewer's personal data, or any biometric information. Only the result, justification identifier, and attendance date are included.

- [ ] *AC6:* Given that the student has notifications enabled, when multiple justifications are evaluated, then the system sends individual notifications for each justification result.
- [ ] *AC7:* Given that the notification is generated, when the operation is completed, then the system emits a platform audit event containing the notification type, the student identifier, the justification identifier, the result, and the notification delivery status.
- [ ] *AC8:* Given normal system conditions, when the justification result is processed, then the notification is generated and queued for delivery within 60 seconds of the evaluation.

Technical notes

- RF 6.3 indicates that the superior provides the justification response automatically and that the support must exist at the time of the response.
- The notification mechanism must be defined by the team. Possible channels include:
    - In-app notification, as the primary channel.
    - Email notification, if configured and approved.
    - Push notification, if the mobile app supports it.
    - SMS or external messaging channels are out of scope unless explicitly approved.
- The notification should be generated as a reaction to the `JustificationApproved` or `JustificationRejected` events from HU-JUS-002.
- The notification must not include:
    - The support document content.
    - The reviewer's personal data beyond what is institutionally allowed.
    - Biometric or sensitive information.

- If the notification delivery fails, the system should:
    - Log the failure.
    - Retry according to a configured policy, or mark it for administrative review.
    - Not block the justification evaluation process.
- The student must be able to consult their notifications even if they missed the initial delivery.
- Notification preferences may be managed by the student through HU-CONF-001 and HU-CONF-002, if those stories are implemented.
- Domain rule RN-36 states that the result of a justification must be notified automatically to the requesting user.
- The team must define the maximum acceptable delay between evaluation and notification delivery. Proposed initial value: 60 seconds.

Responsible service(s): Justification service / Notification service, proposed.

Endpoint(s) implemented:

- `GET /api/v1/notifications/me` — proposed, returns notifications for the authenticated user.
- `GET /api/v1/notifications/{notificationId}` — proposed, returns notification detail.
- `PATCH /api/v1/notifications/{notificationId}/read` — proposed, marks a notification as read.

Events generated:

- `JustificationResultNotificationSent`
- `JustificationResultNotificationFailed`
- `NotificationRead`

Required permissions:

- System-generated notification, not user-triggered.

- Authenticated student/learner may consult and read their own notifications.
- Administrators or Supervisors may review notification delivery failures.

Definition of Done (DoD)

This HU can only be closed when it meets the team's full DoD.

Additional checks specific to this HU:

- [ ] Automatic notification is generated after justification approval.
- [ ] Automatic notification is generated after justification rejection.
- [ ] Notification content includes justification identifier, attendance date, and result.
- [ ] Notification does not include sensitive data such as support content or reviewer personal data.
- [ ] Student can consult notifications in the notification inbox.
- [ ] Notification delivery failure is logged.
- [ ] Retry or administrative review mechanism is tested.
- [ ] Multiple justification evaluations generate individual notifications.
- [ ] Notification is generated within the acceptable delay after evaluation.
- [ ] Audit event is generated for notification delivery.
- [ ] Confirmation and error messages are clear.
- [ ] Traceability to RF 6.3 and Use Case 3 is documented.

Estimation and priority

Table 16

Estimation and priority

Field	Value
Story Points	5
Priority	High
Target sprint	Sprint 2
Dependencies	HU-JUS-002: Justification approval and rejection; HU-CONF-001: Configure system alerts, if applicable; HU-CONF-002: Enable or disable alerts, if applicable

Source. Own elaboration.

HU-HIST

HU-HIST-001: Attendance and absence query by parameters

Story

As an administrator, supervisor, instructor, or student

I want to query attendance and absence records using filters such as date, cohort, environment, instructor, or person

So that I can obtain accurate attendance information according to my role and make timely academic decisions.

Acceptance criteria

Format: "Given [context/initial state], when [user action], then [expected and verifiable result]."

Parameter validation

- [ ] *AC1:* Given that an authenticated user accesses the attendance history module, when the user selects one or more query parameters (date, cohort, environment, instructor, or person), then the system validates each parameter before processing the query.
- [ ] *AC2:* Given that the user selects a date parameter, when the query is executed, then the system validates that the date is valid and applicable to the attendance control context before returning results.
- [ ] *AC3:* Given that the user selects a cohort/ficha as a filter, when the query is executed, then the system validates that the cohort is registered in the system before returning results.
- [ ] *AC4:* Given that the user selects an environment as a filter, when the query is executed, then the system validates that the environment is registered in the system before returning results.
- [ ] *AC5:* Given that the user selects an instructor as a filter, when the query is executed, then the system validates that the instructor/teacher is registered in the

system before returning results.

- [ ] *AC6:* Given that the user selects a person as a filter, when the query is executed, then the system validates that the person is registered in the system before returning results.

Query execution

- [ ] *AC7:* Given that all selected parameters are valid, when the query is executed, then the system returns the attendance and absence records matching the requested filters.
- [ ] *AC8:* Given that the selected parameters are valid but produce no matching records, when the query is executed, then the system shows a clear message indicating that no attendance records were found for the specified filters.
- [ ] *AC9:* Given that the user combines multiple filters, when the query is executed, then the system applies all filters simultaneously and returns only records matching all criteria.

Role-based access

- [ ] *AC10:* Given that the user is a student, when the user queries attendance records, then the system only returns the records belonging to that student, unless a broader permission is explicitly granted.
- [ ] *AC11:* Given that the user is an instructor, when the user queries attendance records, then the system only returns records associated with the cohorts, environments, or training blocks assigned to that instructor.
- [ ] *AC12:* Given that the user is an administrator or supervisor, when the user queries attendance records, then the system returns records according to the administrative scope assigned to that role.
- [ ] *AC13:* Given that an unauthorized user attempts to query attendance records outside their role scope, when the request is processed, then the system denies access and records an unauthorized access attempt.

Results display

- [ ] *AC14:* Given that a query returns results, when the system displays them, then the records include at minimum the following information: user, date, entry time, exit time, attendance status (present, absent, late, justified), cohort, environment, and instructor, where applicable.
- [ ] *AC15:* Given that a query returns a large number of results, when the system displays them, then the results are paginated to maintain performance and usability.

Performance and audit

- [ ] *AC16:* Given normal network conditions, when the user executes an attendance query, then the system responds in less than 5 seconds, according to RNF 1.
- [ ] *AC17:* Given that the query is executed, when the operation is completed, then the system records an audit event containing the user, applied parameters, date, and time.

Technical notes

- RF 5.1 indicates that the system must validate the following parameters before returning results:
    - User is registered in the system.
    - Cohort/ficha is registered in the system.
    - Environment is registered in the system.
    - Instructor/teacher is registered in the system.
    - Date is valid and applicable.
- The query parameters defined by the SRS are: date, cohort/ficha, environment, instructor, and person.
- Role-based filtering must be enforced on the server side:
    - Students: only their own records.
    - Instructors: records from assigned cohorts/environments/training blocks.
    - Administrators/Supervisors: broader administrative scope.

- The attendance status values should align with the domain model defined in `entities-and-rules.md`:
    - Present.
    - Absent.
    - Late.
    - Justified.
- Pagination should be considered for large result sets.
- The system should consider caching strategies for frequently accessed reports, if applicable.
- This story does not cover report export. That is handled in HU-REP-001.
- This story does not cover dashboards. That is handled in HU-REP-002.
- This story does not cover delay/absence-specific queries by parameters. That is handled in HU-REP-004.
- Attendance data must be protected according to RNF 4, and access must be restricted according to the user's role.

Responsible service(s): History service / Attendance query service, proposed.

Endpoint(s) implemented:

- `GET /api/v1/attendance/history` — proposed, with query parameters for date, cohort, environment, instructor, and person.
- `GET /api/v1/attendance/history/me` — proposed, for students querying their own records.

Events generated:

- `AttendanceHistoryQueried` — recommended for audit purposes.
- `UnauthorizedAttendanceQueryAttempt`

Required permissions:

- Authenticated user.
- Role-based scope enforcement:

- Student: own records only.
- Instructor: assigned cohorts, environments, or training blocks.
- Administrator/Supervisor: administrative scope.

Definition of Done (DoD)

This HU can only be closed when it meets the team's full DoD.

Additional checks specific to this HU:

- [ ] Query by date tested.
- [ ] Query by cohort tested.
- [ ] Query by environment tested.
- [ ] Query by instructor tested.
- [ ] Query by person tested.
- [ ] Combined filters tested.
- [ ] Empty result scenario tested.
- [ ] Invalid date scenario tested.
- [ ] Non-existent entity scenario tested.
- [ ] Role-based filtering tested for student, instructor, administrator, and supervisor.
- [ ] Unauthorized access attempt tested.
- [ ] Returned records include required fields.
- [ ] Pagination tested for large result sets.
- [ ] Audit event is generated.
- [ ] Confirmation and error messages are clear.
- [ ] Response time is validated under normal conditions and remains under 5 seconds.
- [ ] Traceability to RF 5.1 is documented.

Estimation and priority

Table 17

Estimation and priority

Field	Value
Story Points	8
Priority	High
Target sprint	Sprint 2
Dependencies	HU-IAM-001: User login; HU-IAM-003: Role-based permission assignment; HU-ATT-001: Facial attendance registration; HU-ACAD-005: Assign conference/time blocks

Source. Own elaboration.

HU-HIST-002: User change history

Story

As an administrator or supervisor

I want to consult the change history of a specific user

So that I can audit the modifications made to that user's data over time and maintain traceability of administrative actions.

Acceptance criteria

Format: "Given [context/initial state], when [user action], then [expected and verifiable result]."

Query validation

- [ ] *AC1:* Given that an authenticated administrator or supervisor accesses the user change history module, when the user provides the parameters to identify the target user, then the system validates that the target user exists and is registered in the application.
- [ ] *AC2:* Given that the target user does not exist in the system, when the administrator queries the change history, then the system rejects the operation and shows a clear validation message.
- [ ] *AC3:* Given that the target user exists but has no registered changes, when the administrator queries the change history, then the system shows a clear message indicating that no changes have been recorded for that user.

History display

- [ ] *AC4:* Given that the target user has registered changes, when the history is displayed, then the system shows each change with at least the following information: date and time of the change, type of change, the user who made the change, previous value (if applicable), and new value (if applicable).
- [ ] *AC5:* Given that the change history contains sensitive data, when the history is displayed, then the system protects sensitive fields (such as passwords or biometric parameters) and does not expose their actual values in the interface or logs.
- [ ] *AC6:* Given that the change history is displayed, when the administrator interacts with the results, then the system supports filtering by target user, date range, type of change, and user who made the change.

Access control

- [ ] *AC7:* Given that the querying user does not have administrator or supervisor permissions, when the user attempts to access the user change history module, then the system denies access and records an unauthorized access attempt.
- [ ] *AC8:* Given that a regular user, instructor, or student attempts to access another user's change history, when the request is processed, then the system denies access and enforces role-based restrictions.

Audit integration

- [ ] *AC9:* Given that a user's data is modified in any other module (user registration, role change, activation/deactivation, facial parameter update, etc.), when the modification is saved, then the system records the change in the user change history automatically.
- [ ] *AC10:* Given that the change history is consulted, when the operation is completed, then the system records an audit event containing the querying user, target user, date, and time.

Performance

- [ ] *AC11:* Given normal network conditions, when the administrator queries the user change history, then the system responds in less than 5 seconds, according to RNF 1.

Technical notes

- RF 5.2 indicates that the system must validate:
    - The querying user is registered in the system.
    - The filtered/target user exists.
- This story depends on audit events being generated by other modules:
    - HU-IAM-001: Login events.
    - HU-IAM-002: User registration and role assignment events.
    - HU-IAM-007: User activation/deactivation events.
    - HU-IAM-010: Role change events.
    - HU-ATT-002: Facial parameter enrollment events.
    - HU-CONF-006: Facial parameter update events.
- The change history must be immutable: once recorded, change entries should not be modified or deleted.
- Sensitive fields such as passwords or biometric parameters must not expose their actual values in the change history. Only the fact that a change occurred should be recorded.
- The change history should support filtering by:
    - Target user.
    - Date range.
    - Type of change.
    - User who made the change.
- Pagination should be considered for users with many changes.
- This story does not cover attendance history. That is handled in HU-HIST-001.

- This story does not cover general audit of all system actions. It focuses specifically on user data changes.
- The team must define the retention period for change history records, considering institutional policies and data protection regulations.

Responsible service(s): attendance-service (HIST-001) / identity-service via platform logging (HIST-002).

Endpoint(s) implemented:

- `GET /api/v1/users/{userId}/change-history` — proposed.
- `GET /api/v1/users/{userId}/change-history` with query parameters for date range, type of change, and author — proposed.

Events generated:

- `UserChangeHistoryQueried` — recommended for audit purposes.
- `UnauthorizedChangeHistoryAccessAttempt`

Required permissions:

- Administrator or Supervisor.
- Regular users must not be able to access other users' change history.

Definition of Done (DoD)

This HU can only be closed when it meets the team's full DoD.

Additional checks specific to this HU:

- [ ] Successful change history query tested.
- [ ] Target user does not exist scenario tested.
- [ ] Target user with no changes scenario tested.
- [ ] Change history includes required fields: date, time, type, author, previous value, new value.
- [ ] Change history is automatically recorded when user data is modified in other modules.
- [ ] Unauthorized access attempt tested.

- [ ] Sensitive fields are protected and not exposed.
- [ ] Change history is immutable.
- [ ] Filtering by date range, type of change, and author tested.
- [ ] Audit event is generated for history queries.
- [ ] Confirmation and error messages are clear.
- [ ] Response time is validated under normal conditions and remains under 5 seconds.
- [ ] Traceability to RF 5.2 is documented.

Estimation and priority

Table 18

Estimation and priority

Field	Value
Story Points	5
Priority	Medium
Target sprint	Sprint 3
Dependencies	HU-IAM-001: User login; HU-IAM-003: Role-based permission assignment; HU-IAM-002: User registration with role assignment; HU-IAM-007: Activate/deactivate users; HU-IAM-010: Role change

Source. Own elaboration.

HU-REP

HU-REP-001: Report export

Story

As an administrator, supervisor, instructor, or authorized user

I want to export attendance reports including assistances, absences, delays, cohorts/groups, instructors, and environments

So that I can obtain structured files for institutional analysis, academic follow-up, and administrative record-keeping according to my role.

Acceptance criteria

Format: "Given [context/initial state], when [user action], then [expected and verifiable result]."

- [ ] *AC1:* Given that an authenticated user accesses the report export module, when the system displays the available report types, then the user can see the following options: assistances, absences, delays, cohorts/groups, instructors/teachers, and environments/classrooms.
- [ ] *AC2:* Given that the user selects a report type, when the export request is submitted, then the system validates that the user's role is authorized to export that specific report type before processing the request.
- [ ] *AC3:* Given that the user's role is authorized for the selected report type, when the export is processed, then the system generates the report file and provides it to the user for download.
- [ ] *AC4:* Given that the user's role is not authorized for the selected report type, when the export is requested, then the system denies the request, shows a clear permission error message, and records an unauthorized access attempt.
- [ ] *AC5:* Given that the report generation completes successfully, when the file is downloaded, then the exported data matches the current authorized scope of the user and includes the report type, generation date, and user who requested the export.

- [ ] *AC6:* Given that the report contains no data matching the user's scope, when the export is requested, then the system shows a clear message indicating that no data is available for export under the current criteria.
- [ ] *AC7:* Given that a student attempts to export reports outside their personal scope, when the request is processed, then the system denies the request and enforces role-based restrictions.
- [ ] *AC8:* Given normal network conditions, when the user submits a standard export request, then the system responds in less than 10 seconds for typical data volumes.
- [ ] *AC9:* Given that the export request is processed, when the operation is completed, then the system records an audit event containing the user, report type, date, time, and result.

Technical notes

- RF 7.1 indicates that the system must validate the user's role and associate it with the export types they are allowed to request.
- The report types defined by the SRS are:
    - Attendances (orig. ES "Asistencias").
    - Absences (orig. ES "Faltas").
    - Delays (orig. ES "Retardos").
    - Cohorts/groups (orig. ES "Fichas/Grupos").
    - Instructors/teachers (orig. ES "Instructores/Profesores").
    - Environments/classrooms (orig. ES "Ambientes/Salón").
- Role-based access matrix for reports should be defined by the team. Proposed initial matrix:
    - Student: personal attendance history only.
    - Instructor: assistances, delays, and absences for assigned cohorts.
    - Administrator/Supervisor: all report types.
- Export formats must be defined by the team. Proposed initial formats:

    - CSV for data analysis.
    - PDF for institutional documents.
- For large datasets, the system should consider:
    - Pagination or streaming generation.
    - Asynchronous export with notification when ready.
    - A maximum export size limit.
- The exported file should include:
    - Report type.
    - Generation date and time.
    - User who requested the export.
    - Applied filters, if any.
- Exported data must respect the user's role scope and must not include unauthorized records.
- This story does not cover date-range filters. That is handled in HU-REP-003.
- This story does not cover the analytical dashboard. That is handled in HU-REP-002.
- This story does not cover delay/absence-specific queries by parameters. That is handled in HU-REP-004.
- Performance for large exports may exceed the standard 5-second response time. The team should define whether asynchronous export is required.
- Sensitive data included in exports must be protected according to RNF 4.

Responsible service(s): Report service / Export service, proposed.

Endpoint(s) implemented:

- `GET /api/v1/reports/types` — proposed, returns report types available for the user's role.
- `POST /api/v1/reports/export` — proposed, generates and returns the report file.
- `GET /api/v1/reports/export/{exportId}` — proposed, retrieves an asynchronous export result.

Events generated:

- `ReportExportRequested`
- `ReportExportCompleted`
- `ReportExportFailed`
- `UnauthorizedReportAccessAttempt`

Required permissions:

- Authenticated user.
- Role-based scope enforcement:
    - Student: personal records only.
    - Instructor: assigned cohorts/environments.
    - Administrator/Supervisor: full institutional scope.

Definition of Done (DoD)

This HU can only be closed when it meets the team's full DoD.

Additional checks specific to this HU:

- [ ] All six report types are available according to role permissions.
- [ ] Role-based validation tested for each report type.
- [ ] Successful export tested for authorized roles.
- [ ] Unauthorized export attempt tested and denied.
- [ ] Empty result scenario tested.
- [ ] Student scope restriction tested.
- [ ] Exported file includes report type, generation date, and requesting user.
- [ ] Audit event is generated for export requests.
- [ ] Confirmation and error messages are clear.
- [ ] Response time is validated for typical data volumes.
- [ ] Large dataset behavior is defined and tested.
- [ ] Traceability to RF 7.1 is documented.

Estimation and priority

Table 19

Estimation and priority

Field	Value
Story Points	8
Priority	Medium
Target sprint	Sprint 3
Dependencies	HU-IAM-001: User login; HU-IAM-003: Role-based permission assignment; HU-HIST-001: Attendance and absence query by parameters

Source. Own elaboration.

HU-REP-002: Role-based analytical dashboard

Story

As an administrator, supervisor, instructor, or student

I want to access an analytical dashboard corresponding to my role

So that I can visualize relevant attendance indicators and make timely decisions according to my responsibilities.

Acceptance criteria

Format: "Given [context/initial state], when [user action], then [expected and verifiable result]."

- [ ] *AC1:* Given that an authenticated user accesses the dashboard module, when the system processes the request, then the system validates the user's role and displays the dashboard corresponding to that role.
- [ ] *AC2:* Given that the user is a student, when the dashboard is displayed, then the system shows personal attendance information such as attendance percentage, absences, delays, and justification status.
- [ ] *AC3:* Given that the user is an instructor, when the dashboard is displayed, then the system shows attendance information for the cohorts, environments, or training

blocks assigned to that instructor.

- [ ] *AC4:* Given that the user is an administrator or supervisor, when the dashboard is displayed, then the system shows institutional or administrative-scope attendance indicators.
- [ ] *AC5:* Given that the user's role does not have an associated dashboard, when the user accesses the dashboard module, then the system shows a clear message indicating that no dashboard is available for their role.
- [ ] *AC6:* Given that the dashboard is displayed, when the user interacts with the visual components, then the information is updated according to the user's scope without exposing unauthorized data.
- [ ] *AC7:* Given that an unauthorized user attempts to access a dashboard outside their role, when the request is processed, then the system denies access and records an unauthorized access attempt.
- [ ] *AC8:* Given normal network conditions, when the user accesses the dashboard, then the system responds in less than 5 seconds, according to RNF 1.
- [ ] *AC9:* Given that the dashboard is accessed, when the operation is completed, then the system records an audit event containing the user, role, date, and time.

Technical notes

- RF 7.2 indicates that the system must validate the user's role and associate it with the corresponding dashboard.
- The dashboard content must be defined per role. Proposed initial content:
    - *Student dashboard:*
        - Attendance percentage.
        - Total absences.
        - Total delays.
        - Justification status.
        - Recent attendance records.

    - *Instructor dashboard:*
        - Attendance summary for assigned cohorts.
        - Students with most absences or delays.
        - Recent attendance records.
        - Pending justifications, if applicable.
    - *Administrator/Supervisor dashboard:*
        - Institutional attendance summary.
        - Attendance by cohort.
        - Attendance by environment.
        - Users with most absences or delays.
        - System alerts, if applicable.
- The dashboard should be responsive and compatible with web and mobile views, according to RNF 3.
- The dashboard should consider:
    - Real-time or near-real-time updates, if feasible.
    - Clear visual indicators (charts, counters, status badges).
    - Accessibility best practices, according to RNF 2.
- Role-based filtering must be enforced on the server side. Client-side filtering alone is not sufficient.
- The dashboard should use cached or aggregated data where possible to improve performance.
- This story does not cover report export. That is handled in HU-REP-001.
- This story does not cover date-range reports. That is handled in HU-REP-003.
- This story does not cover delay/absence-specific queries. That is handled in HU-REP-004.
- The team must define the specific KPIs and visual components for each role.

Responsible service(s): Report service / Dashboard service, proposed.

Endpoint(s) implemented:

- `GET /api/v1/dashboard` — proposed, returns dashboard data based on user role.
- `GET /api/v1/dashboard/student` — proposed.
- `GET /api/v1/dashboard/instructor` — proposed.
- `GET /api/v1/dashboard/admin` — proposed.

Events generated:

- `DashboardAccessed`
- `UnauthorizedDashboardAccessAttempt`

Required permissions:

- Authenticated user.
- Role-based scope enforcement:
    - Student: personal dashboard.
    - Instructor: assigned cohorts/environments dashboard.
    - Administrator/Supervisor: institutional dashboard.

Definition of Done (DoD)

This HU can only be closed when it meets the team's full DoD.

Additional checks specific to this HU:

- [ ] Dashboard is displayed correctly for each role.
- [ ] Student dashboard shows personal attendance information.
- [ ] Instructor dashboard shows assigned cohorts information.
- [ ] Administrator/Supervisor dashboard shows institutional indicators.
- [ ] Role without dashboard scenario tested.
- [ ] Unauthorized dashboard access attempt tested.
- [ ] Dashboard data respects user scope.
- [ ] Dashboard is responsive on web and mobile views.
- [ ] Audit event is generated for dashboard access.

- [ ] Confirmation and error messages are clear.
- [ ] Response time is validated under normal conditions and remains under 5 seconds.
- [ ] Traceability to RF 7.2 is documented.

Estimation and priority

Table 20

Estimation and priority

Field	Value
Story Points	8
Priority	High
Target sprint	Sprint 3
Dependencies	HU-IAM-001: User login; HU-IAM-003: Role-based permission assignment; HU-HIST-001: Attendance and absence query by parameters; HU-ATT-001: Facial attendance registration

Source. Own elaboration.

HU-REP-003: Date-range reports

Story

As an administrator, supervisor, instructor, or authorized user

I want to export reports filtered by a date range

So that I can analyze attendance information for specific periods such as weeks, months, or academic terms according to my role.

Acceptance criteria

Format: "Given [context/initial state], when [user action], then [expected and verifiable result]."

- [ ] *AC1:* Given that an authenticated user accesses the date-range report module, when the user selects start date, end date, and report type, then the system validates the input parameters before processing the request.
- [ ] *AC2:* Given that the selected date range is valid and the user's role is authorized, when the export request is submitted, then the system generates the report filtered by

the specified date range and provides it for download.

- [ ] *AC3:* Given that the start date is later than the end date, when the request is submitted, then the system rejects the operation and shows a clear validation message.
- [ ] *AC4:* Given that the selected date range exceeds the maximum allowed period, when the request is submitted, then the system rejects the operation and shows a clear message indicating the maximum allowed period.
- [ ] *AC5:* Given that the user's role is not authorized for the selected report type, when the export is requested, then the system denies the request, shows a permission error, and records an unauthorized access attempt.
- [ ] *AC6:* Given that no data exists within the selected date range, when the export is requested, then the system shows a clear message indicating that no data is available for the specified period.
- [ ] *AC7:* Given that the date-range report is generated, when the file is downloaded, then the exported data respects the user's role scope and includes the applied date range, report type, generation date, and requesting user.
- [ ] *AC8:* Given normal network conditions, when the user submits a standard date-range export request, then the system responds in less than 10 seconds for typical data volumes.
- [ ] *AC9:* Given that the date-range export is processed, when the operation is completed, then the system records an audit event containing the user, report type, date range, date, time, and result.

Technical notes

- RF 7.3 indicates that the system must validate the user's role and associate it with the export types they are allowed to request, similar to RF 7.1 but with date-range filtering.
- The report types available for date-range export should align with HU-REP-001:
    - Attendances (orig. ES "Asistencias").

    - Absences (orig. ES "Faltas").
    - Delays (orig. ES "Retardos").
    - Cohorts/groups (orig. ES "Fichas/Grupos").
    - Instructors/teachers (orig. ES "Instructores/Profesores").
    - Environments/classrooms (orig. ES "Ambientes/Salón").
- The team must define:
    - Maximum allowed date range (for example, 1 year).
    - Whether date-range export supports combined filters (cohort, environment, instructor).
    - Whether asynchronous export is required for large date ranges.
- The system should validate:
    - Start date is not later than end date.
    - Date range does not exceed the maximum allowed period.
    - Dates are valid and applicable to the attendance control context.
- The exported file should include:
    - Applied date range.
    - Report type.
    - Generation date and time.
    - User who requested the export.
- This story builds on HU-REP-001. The team should consider reusing the export generation logic.
- This story does not cover the analytical dashboard. That is handled in HU-REP-002.
- This story does not cover delay/absence-specific queries by parameters. That is handled in HU-REP-004.
- Performance for large date ranges may require asynchronous export with notification when ready.
- Sensitive data included in exports must be protected according to RNF 4.

Responsible service(s): Report service / Export service, proposed.

Endpoint(s) implemented:

- `POST /api/v1/reports/export/by-date-range` — proposed.
- `GET /api/v1/reports/export/{exportId}` — proposed, retrieves an asynchronous export result.

Events generated:

- `DateRangeReportRequested`
- `DateRangeReportCompleted`
- `DateRangeReportFailed`
- `UnauthorizedDateRangeReportAccessAttempt`

Required permissions:

- Authenticated user.
- Role-based scope enforcement:
    - Student: personal records only.
    - Instructor: assigned cohorts/environments.
    - Administrator/Supervisor: full institutional scope.

Definition of Done (DoD)

This HU can only be closed when it meets the team's full DoD.

Additional checks specific to this HU:

- [ ] Successful date-range export tested.
- [ ] Start date later than end date scenario tested.
- [ ] Maximum date range exceeded scenario tested.
- [ ] Unauthorized report type scenario tested.
- [ ] Empty date range result scenario tested.
- [ ] Role-based scope enforcement tested.
- [ ] Exported file includes applied date range, report type, and requesting user.
- [ ] Audit event is generated for date-range export requests.

- [ ] Confirmation and error messages are clear.
- [ ] Response time is validated for typical data volumes.
- [ ] Large date range behavior is defined and tested.
- [ ] Traceability to RF 7.3 is documented.

Estimation and priority

Table 21

Estimation and priority

Field	Value
Story Points	5
Priority	Medium
Target sprint	Sprint 3
Dependencies	HU-REP-001: Report export; HU-IAM-003: Role-based permission assignment; HU-HIST-001: Attendance and absence query by parameters

Source. Own elaboration.

HU-REP-004: Delay and absence query by parameters

Story

As an administrator, supervisor, instructor, or authorized user

I want to query delays and absences using parameters such as user, date, cohort/group, environment/classroom, and instructor

So that I can identify attendance issues and take timely corrective actions according to my role.

Acceptance criteria

Format: "Given [context/initial state], when [user action], then [expected and verifiable result]."

- [ ] *AC1:* Given that an authenticated user accesses the delay/absence query module, when the user selects one or more parameters (user, date, cohort/group, environment/classroom, instructor), then the system validates the parameters before processing the query.

- [ ] *AC2:* Given that the selected parameters are valid and the user's role is authorized, when the query is executed, then the system returns the delays and absences matching the requested filters.
- [ ] *AC3:* Given that the user's role is not authorized for the requested query scope, when the query is executed, then the system denies access, shows a permission error, and records an unauthorized access attempt.
- [ ] *AC4:* Given that the selected parameters produce no matching records, when the query is executed, then the system shows a clear message indicating that no delays or absences were found for the specified filters.
- [ ] *AC5:* Given that a student queries delays and absences, when the request is processed, then the system only returns records belonging to that student.
- [ ] *AC6:* Given that an instructor queries delays and absences, when the request is processed, then the system only returns records associated with the cohorts, environments, or training blocks assigned to that instructor.
- [ ] *AC7:* Given that the query returns results, when the system displays them, then each record includes at minimum the following information: user, date, type (delay or absence), cohort, environment, instructor, and justification status where applicable.
- [ ] *AC8:* Given normal network conditions, when the user executes a delay/absence query, then the system responds in less than 5 seconds, according to RNF 1.
- [ ] *AC9:* Given that the query is executed, when the operation is completed, then the system records an audit event containing the user, applied parameters, date, and time.

Technical notes

- RF 7.4 indicates that the system must validate the user's role and associate it with the delay/absence queries they are allowed to request.
- The query parameters defined by the SRS are:
    - User references.
    - Dates.

    - Cohorts/groups.
    - Environments/classrooms.
    - Instructors/teachers.
- The query should support combined filters for more precise results.
- Role-based filtering must be enforced on the server side:
    - Students: only their own records.
    - Instructors: records from assigned cohorts/environments/training blocks.
    - Administrators/Supervisors: broader administrative scope.
- The query results should include justification status to help users identify which delays/absences have been justified.
- Pagination should be considered for large result sets.
- The team should consider:
    - Sorting options (by date, user, type).
    - Export option for query results, if applicable.
    - Integration with HU-REP-001 for exporting query results.
- This story does not cover the analytical dashboard. That is handled in HU-REP-002.
- This story does not cover general attendance/absence history queries. That is handled in HU-HIST-001.
- This story focuses specifically on delays and absences, not on all attendance records.
- Attendance data must be protected according to RNF 4, and access must be restricted according to the user's role.

Responsible service(s): Report service / Delay-Absence query service, proposed.

Endpoint(s) implemented:

- `GET /api/v1/reports/delays-absences` — proposed, with query parameters for user, date, cohort, environment, and instructor.
- `GET /api/v1/reports/delays-absences/me` — proposed, for students querying their own records.

Events generated:

- `DelayAbsenceQueryExecuted` — recommended for audit purposes.
- `UnauthorizedDelayAbsenceQueryAttempt`

Required permissions:

- Authenticated user.
- Role-based scope enforcement:
    - Student: own records only.
    - Instructor: assigned cohorts, environments, or training blocks.
    - Administrator/Supervisor: administrative scope.

Definition of Done (DoD)

This HU can only be closed when it meets the team's full DoD.

Additional checks specific to this HU:

- [ ] Query by user tested.
- [ ] Query by date tested.
- [ ] Query by cohort/group tested.
- [ ] Query by environment/classroom tested.
- [ ] Query by instructor tested.
- [ ] Combined filters tested.
- [ ] Empty result scenario tested.
- [ ] Role-based filtering tested for student, instructor, administrator, and supervisor.
- [ ] Unauthorized access attempt tested.
- [ ] Returned records include required fields: user, date, type, cohort, environment, instructor, justification status.
- [ ] Pagination tested for large result sets.
- [ ] Audit event is generated.
- [ ] Confirmation and error messages are clear.
- [ ] Response time is validated under normal conditions and remains under 5 seconds.

- [ ] Traceability to RF 7.4 is documented.

Estimation and priority

Table 22

Estimation and priority

Field	Value
Story Points	5
Priority	High
Target sprint	Sprint 3
Dependencies	HU-IAM-001: User login; HU-IAM-003: Role-based permission assignment; HU-HIST-001: Attendance and absence query by parameters; HU-CONF-008: Delay and justification parametrization

Source. Own elaboration.

HU-CONF

HU-CONF-001: Configure system alerts

Story

As an authenticated user

I want to configure the alerts and notifications of the application

So that I can customize how the system informs me about relevant events such as attendance records, justifications, and system alerts.

Acceptance criteria

Format: "Given [context/initial state], when [user action], then [expected and verifiable result]."

- [ ] *AC1:* Given that the user is registered and has logged into the system, when the user accesses the alert configuration module, then the system displays the available alert configuration options.
- [ ] *AC2:* Given that the user modifies one or more alert configuration options, when the changes are saved, then the system stores the new configuration and reflects the changes in the system's behavior.
- [ ] *AC3:* Given that the user is not registered or has not logged into the system, when the user attempts to access the alert configuration module, then the system denies access and shows a clear authentication error message.
- [ ] *AC4:* Given that the alert configuration is saved successfully, when the system processes the changes, then the system shows a confirmation message indicating that the configuration was updated.
- [ ] *AC5:* Given that the alert configuration fails to save due to a system error, when the operation is processed, then the system shows a clear error message and does not apply the changes.
- [ ] *AC6:* Given that the user's alert configuration is updated, when a relevant event occurs in the system, then the system sends notifications according to the user's

configured preferences.

- [ ] *AC7:* Given normal network conditions, when the user saves the alert configuration, then the system responds in less than 5 seconds, according to RNF 1.
- [ ] *AC8:* Given that the alert configuration is updated, when the operation is completed, then the system records an audit event containing the user, date, and time.

Technical notes

- RF 4.1 indicates that the system must validate that the user is registered and has logged into the system before allowing alert configuration.
- This story covers the general alert configuration module. Specific sub-options such as enabling/disabling alerts and changing notification tones are handled in separate stories (HU-CONF-002 and HU-CONF-003).
- The alert configuration should be stored per user to allow personalization.
- The system should consider the following types of alerts:
    - Attendance records (entry/exit).
    - Justification results (approved/rejected).
    - System alerts (device failures, maintenance).
    - Academic alerts (delays, absences).
- The alert configuration should be compatible with the notification delivery mechanism defined in HU-JUS-003.
- The system must validate that the user is authenticated before processing any configuration change.
- Configuration changes must be auditable for traceability.
- This story does not cover IoT device failure alerts. That is handled in HU-CONF-007.
- The team must define the default alert configuration for new users.

Responsible service(s): Configuration service / Notification service, proposed.

Endpoint(s) implemented:

- `GET /api/v1/config/alerts` — proposed, returns the user's current alert configuration.
- `PUT /api/v1/config/alerts` — proposed, updates the user's alert configuration.

Events generated:

- `AlertConfigurationUpdated`
- `AlertConfigurationFailed`

Required permissions:

- Authenticated user (any role).
- Configuration is personal and applies only to the authenticated user's notifications.

Definition of Done (DoD)

This HU can only be closed when it meets the team's full DoD.

Additional checks specific to this HU:

- [ ] Alert configuration module is accessible to authenticated users.
- [ ] Successful configuration update tested.
- [ ] Unauthenticated access attempt tested.
- [ ] Confirmation and error messages are clear.
- [ ] Configuration is stored per user.
- [ ] Notifications are sent according to configured preferences.
- [ ] Audit event is generated.
- [ ] Response time is validated under normal conditions and remains under 5 seconds.
- [ ] Traceability to RF 4.1 is documented.

Estimation and priority

Table 23

Estimation and priority

Field	Value
Story Points	3
Priority	Medium

Field	Value
Target sprint	Sprint 3
Dependencies	HU-IAM-001: User login

Source. Own elaboration.

HU-CONF-002: Enable or disable alerts

Story

As an authenticated user

I want to enable or disable specific alerts and notifications

So that I can control which notifications I receive and reduce unnecessary interruptions.

Acceptance criteria

Format: "Given [context/initial state], when [user action], then [expected and verifiable result]."

- [ ] *AC1:* Given that the user is registered and has logged into the system, when the user accesses the alert enable/disable module, then the system displays the list of available alerts with their current enabled/disabled status.
- [ ] *AC2:* Given that the user selects an alert to enable, when the change is saved, then the system enables the selected alert and reflects the change in the system's behavior.
- [ ] *AC3:* Given that the user selects an alert to disable, when the change is saved, then the system disables the selected alert and limits the notifications the user receives for that alert type.
- [ ] *AC4:* Given that the user is not registered or has not logged into the system, when the user attempts to enable or disable alerts, then the system denies access and shows a clear authentication error message.
- [ ] *AC5:* Given that the alert status is changed successfully, when the system processes the change, then the system shows a confirmation message indicating that the alert was enabled or disabled.

- [ ] *AC6:* Given that an alert is disabled, when the corresponding event occurs in the system, then the system does not send notifications for that alert type to the user.
- [ ] *AC7:* Given that an alert is enabled, when the corresponding event occurs in the system, then the system sends notifications for that alert type to the user according to their configured preferences.
- [ ] *AC8:* Given normal network conditions, when the user enables or disables an alert, then the system responds in less than 5 seconds, according to RNF 1.
- [ ] *AC9:* Given that the alert status is changed, when the operation is completed, then the system records an audit event containing the user, alert type, new status, date, and time.

Technical notes

- RF 4.1.1 indicates that the system must validate that the user is registered and has logged into the system before allowing alert enable/disable operations.
- This story specifically covers the enable/disable toggle for individual alerts, complementing the general configuration module in HU-CONF-001.
- The alert types that can be enabled or disabled should align with the alert categories defined in HU-CONF-001:
    - Attendance records (entry/exit).
    - Justification results (approved/rejected).
    - System alerts (device failures, maintenance).
    - Academic alerts (delays, absences).
- The system should consider a master toggle to enable or disable all alerts at once, as well as individual toggles for each alert type.
- When an alert is disabled, the system must not send notifications for that alert type, but should still log the event for audit purposes.
- The alert status should be stored per user to allow personalization.

- The team must define whether critical alerts (such as security incidents) can be disabled by users or should always remain enabled.
- This story does not cover changing notification tones. That is handled in HU-CONF-003.

Responsible service(s): Configuration service / Notification service, proposed.

Endpoint(s) implemented:

- `GET /api/v1/config/alerts/status` — proposed, returns the enabled/disabled status of each alert.
- `PATCH /api/v1/config/alerts/{alertType}/status` — proposed, enables or disables a specific alert.

Events generated:

- `AlertEnabled`
- `AlertDisabled`
- `AlertStatusChangeFailed`

Required permissions:

- Authenticated user (any role).
- Configuration is personal and applies only to the authenticated user's notifications.

Definition of Done (DoD)

This HU can only be closed when it meets the team's full DoD.

Additional checks specific to this HU:

- [ ] Alert enable/disable module is accessible to authenticated users.
- [ ] Successful alert enable tested.
- [ ] Successful alert disable tested.
- [ ] Unauthenticated access attempt tested.
- [ ] Notifications are not sent for disabled alerts.
- [ ] Notifications are sent for enabled alerts.
- [ ] Master toggle (if implemented) tested.

- [ ] Audit event is generated.
- [ ] Confirmation and error messages are clear.
- [ ] Response time is validated under normal conditions and remains under 5 seconds.
- [ ] Traceability to RF 4.1.1 is documented.

Estimation and priority

Table 24

Estimation and priority

Field	Value
Story Points	2
Priority	Low
Target sprint	Sprint 3
Dependencies	HU-IAM-001: User login; HU-CONF-001: Configure system alerts

Source. Own elaboration.

HU-CONF-003: Change notification tone

Story

As an authenticated user

I want to change the tone or sound of the application's notifications

So that I can customize the audio feedback I receive when the system alerts me about relevant events.

Acceptance criteria

Format: "Given [context/initial state], when [user action], then [expected and verifiable result]."

- [ ] *AC1:* Given that the user is registered and has logged into the system, when the user accesses the notification tone configuration module, then the system displays the list of available notification tones.
- [ ] *AC2:* Given that the user selects a notification tone from the available options, when the change is saved, then the system stores the selected tone and applies it to the

user's notifications.

- [ ] *AC3:* Given that the user selects a notification tone, when the user wants to preview the sound, then the system allows playing a preview of the selected tone before saving.
- [ ] *AC4:* Given that the user is not registered or has not logged into the system, when the user attempts to change the notification tone, then the system denies access and shows a clear authentication error message.
- [ ] *AC5:* Given that the notification tone is changed successfully, when the system processes the change, then the system shows a confirmation message indicating that the tone was updated.
- [ ] *AC6:* Given that a notification is triggered after the tone change, when the notification is delivered, then the system plays the newly selected tone.
- [ ] *AC7:* Given that the user does not select a tone or cancels the operation, when the module is closed, then the system retains the previous tone without applying changes.
- [ ] *AC8:* Given normal network conditions, when the user changes the notification tone, then the system responds in less than 5 seconds, according to RNF 1.
- [ ] *AC9:* Given that the notification tone is changed, when the operation is completed, then the system records an audit event containing the user, selected tone, date, and time.

Technical notes

- RF 4.1.2 indicates that the system must validate that the user is registered and has logged into the system before allowing notification tone changes.
- This story specifically covers the audio customization of notifications, complementing the general configuration module in HU-CONF-001.
- The system should provide a predefined set of notification tones for the user to choose from. The team must define the available tones.
- The system should consider:

    - A default tone for new users.
    - A "silent" or "vibration only" option, if applicable.
    - Preview functionality before saving.
- The notification tone setting should be stored per user to allow personalization.
- On mobile devices, the application must request appropriate permissions to play sounds.
- On web browsers, the system should handle browser autoplay policies that may restrict audio playback.
- The tone change should apply to all notification types unless the team decides to allow per-alert-type tone configuration.
- This story does not cover enabling/disabling alerts. That is handled in HU-CONF-002.

Responsible service(s): Configuration service / Notification service, proposed.

Endpoint(s) implemented:

- `GET /api/v1/config/notification-tone` — proposed, returns the user's current notification tone.
- `PUT /api/v1/config/notification-tone` — proposed, updates the user's notification tone.

Events generated:

- `NotificationToneUpdated`
- `NotificationToneChangeFailed`

Required permissions:

- Authenticated user (any role).
- Configuration is personal and applies only to the authenticated user's notifications.

Definition of Done (DoD)

This HU can only be closed when it meets the team's full DoD.

Additional checks specific to this HU:

- [ ] Notification tone module is accessible to authenticated users.
- [ ] Available tones are displayed correctly.
- [ ] Successful tone change tested.
- [ ] Preview functionality tested.
- [ ] Unauthenticated access attempt tested.
- [ ] New tone is played on subsequent notifications.
- [ ] Cancel operation retains previous tone.
- [ ] Mobile permissions handling tested.
- [ ] Web browser autoplay policy handling tested.
- [ ] Audit event is generated.
- [ ] Confirmation and error messages are clear.
- [ ] Response time is validated under normal conditions and remains under 5 seconds.
- [ ] Traceability to RF 4.1.2 is documented.

Estimation and priority

Table 25

Estimation and priority

Field	Value
Story Points	2
Priority	Low
Target sprint	Sprint 3
Dependencies	HU-IAM-001: User login; HU-CONF-001: Configure system alerts

Source. Own elaboration.

HU-CONF-004: Configure application language

Story

As an authenticated user

I want to configure the language of the application

So that I can interact with the system in my preferred language and improve my user experience.

Acceptance criteria

Format: "Given [context/initial state], when [user action], then [expected and verifiable result]."

- [ ] *AC1:* Given that the user is registered and has logged into the system, when the user accesses the language configuration module, then the system displays the list of available languages.
- [ ] *AC2:* Given that the user selects a language from the available options, when the change is saved, then the system stores the selected language and applies it to the user's interface.
- [ ] *AC3:* Given that the language is changed successfully, when the system processes the change, then the system shows a confirmation message in the newly selected language.
- [ ] *AC4:* Given that the user is not registered or has not logged into the system, when the user attempts to change the language, then the system denies access and shows a clear authentication error message.
- [ ] *AC5:* Given that the user selects a language, when subsequent screens and messages are displayed, then all interface texts, labels, and messages are shown in the selected language.
- [ ] *AC6:* Given that the user does not select a language or cancels the operation, when the module is closed, then the system retains the previous language without applying changes.
- [ ] *AC7:* Given that a new user registers in the system, when no language preference has been set, then the system uses the default language defined by the team.
- [ ] *AC8:* Given normal network conditions, when the user changes the language, then the system responds in less than 5 seconds, according to RNF 1.
- [ ] *AC9:* Given that the language is changed, when the operation is completed, then the system records an audit event containing the user, selected language, date, and

time.

Technical notes

- RF 4.2 indicates that the system must validate that the user is registered and has logged into the system before allowing language configuration.
- This story supports the non-functional requirement RNF 2 (Disponibilidad), which states that the interface should support different languages.
- The team must define the available languages. Proposed initial languages:
    - Spanish (default).
    - English.
- The system should consider implementing internationalization (i18n) best practices:
    - Externalize all user-facing strings into translation files.
    - Use locale-based formatting for dates, times, and numbers.
    - Support right-to-left (RTL) languages if applicable in the future.
- The language preference should be stored per user to allow personalization.
- The system should apply the language change immediately without requiring a logout/login cycle, if feasible.
- Error messages, validation messages, and notification texts should also be translated.
- The system should consider a fallback language (default) in case a translation is missing.
- This story does not cover color palette configuration. That is handled in HU-CONF-005.
- The team must define whether the language configuration applies only to the web interface or also to mobile notifications and emails.

Responsible service(s): Configuration service / Internationalization service, proposed.

Endpoint(s) implemented:

- `GET /api/v1/config/language` — proposed, returns the user's current language.
- `PUT /api/v1/config/language` — proposed, updates the user's language.

- `GET /api/v1/config/available-languages` — proposed, returns the list of available languages.

Events generated:

- `LanguageUpdated`
- `LanguageChangeFailed`

Required permissions:

- Authenticated user (any role).
- Configuration is personal and applies only to the authenticated user's interface.

Definition of Done (DoD)

This HU can only be closed when it meets the team's full DoD.

Additional checks specific to this HU:

- [ ] Language configuration module is accessible to authenticated users.
- [ ] Available languages are displayed correctly.
- [ ] Successful language change tested.
- [ ] Interface texts are displayed in the selected language.
- [ ] Confirmation message is shown in the new language.
- [ ] Unauthenticated access attempt tested.
- [ ] Cancel operation retains previous language.
- [ ] Default language is applied for new users.
- [ ] Error and validation messages are translated.
- [ ] Date and time formatting adapts to the selected locale.
- [ ] Audit event is generated.
- [ ] Response time is validated under normal conditions and remains under 5 seconds.
- [ ] Traceability to RF 4.2 is documented.

Estimation and priority

Table 26

Estimation and priority

Field	Value
Story Points	3
Priority	Medium
Target sprint	Sprint 3
Dependencies	HU-IAM-001: User login

Source. Own elaboration.

HU-CONF-005: Update color palettes

Story

As an administrator

I want to configure and add color palettes for the application interface

So that the institution's visual identity is reflected in the system and the user experience is customized according to institutional criteria.

Acceptance criteria

Format: "Given [context/initial state], when [user action], then [expected and verifiable result]."

- [ ] *AC1:* Given that an authenticated administrator accesses the color palette configuration module, when the system loads the module, then it displays the current color palettes and the option to add or modify palettes.
- [ ] *AC2:* Given that the administrator selects or creates a new color palette, when the changes are saved, then the system validates the color values, stores the new palette, and reflects the changes in the application's visual behavior.
- [ ] *AC3:* Given that the administrator enters invalid color values (malformed codes, out-of-range values), when the save operation is submitted, then the system rejects the operation and shows specific validation messages.
- [ ] *AC4:* Given that the user attempting to configure color palettes does not have administrator permissions, when the request is processed, then the system denies

access and records an unauthorized access attempt.

- [ ] *AC5:* Given that the color palette is updated successfully, when the system applies the changes, then subsequent screens and interface components reflect the new color scheme.
- [ ] *AC6:* Given that the administrator cancels the operation without saving, when the module is closed, then the system retains the previous color palette without applying changes.
- [ ] *AC7:* Given normal network conditions, when the administrator saves the color palette configuration, then the system responds in less than 5 seconds, according to RNF 1.
- [ ] *AC8:* Given that the color palette is updated, when the operation is completed, then the system records an audit event containing the administrator, date, and time.

Technical notes

- RF 4.3 indicates that the system must validate that the user is registered and has logged into the system before allowing color palette configuration.
- RF 4.3 specifies that this is an *administrator-only* operation. Regular users, instructors, and students must not be able to modify color palettes.
- The color palette configuration should support standard color formats such as hexadecimal codes (e.g., `#RRGGBB`) or RGB values.
- The system should consider:
    - A preview functionality before saving changes.
    - Contrast validation to ensure accessibility compliance (WCAG guidelines).
    - A default palette that is applied if no custom palette is configured.
- The color palette changes should apply globally across the application or per institutional scope, depending on the team's decision.

- The team must define whether multiple palettes can coexist and how they are applied (by institution, by role, or globally).
- This story does not cover language configuration. That is handled in HU-CONF-004.
- This story does not cover alert configuration. That is handled in HU-CONF-001.

Responsible service(s): Configuration service / UI theme service, proposed.

Endpoint(s) implemented:

- `GET /api/v1/config/color-palettes` — proposed, returns current color palettes.
- `POST /api/v1/config/color-palettes` — proposed, adds a new color palette.
- `PUT /api/v1/config/color-palettes/{paletteId}` — proposed, updates an existing palette.

Events generated:

- `ColorPaletteUpdated`
- `ColorPaletteChangeFailed`
- `UnauthorizedPaletteAccessAttempt`

Required permissions:

- Administrator only.
- Regular users, instructors, and students must not access this configuration.

Definition of Done (DoD)

This HU can only be closed when it meets the team's full DoD.

Additional checks specific to this HU:

- [ ] Color palette module is accessible only to administrators.
- [ ] Successful palette creation tested.
- [ ] Successful palette update tested.
- [ ] Invalid color values scenario tested.
- [ ] Unauthorized access attempt tested.
- [ ] Preview functionality tested (if implemented).
- [ ] Interface reflects new color scheme after saving.

- [ ] Cancel operation retains previous palette.
- [ ] Audit event is generated.
- [ ] Confirmation and error messages are clear.
- [ ] Response time is validated under normal conditions and remains under 5 seconds.
- [ ] Traceability to RF 4.3 is documented.

Estimation and priority

Table 27

Estimation and priority

Field	Value
Story Points	3
Priority	Medium
Target sprint	Sprint 3
Dependencies	HU-IAM-001: User login; HU-IAM-003: Role-based permission assignment

Source. Own elaboration.

HU-CONF-006: Update user facial parameters

Story

As an authenticated user or administrator

I want to update my facial parameters in the system

So that my facial recognition data remains accurate and up-to-date for reliable attendance registration.

Acceptance criteria

Format: "Given [context/initial state], when [user action], then [expected and verifiable result]."

- [ ] *AC1:* Given that the user is registered and has logged into the system, when the user requests a facial parameter update, then the system activates the camera and initiates the facial capture process.

- [ ] *AC2:* Given that the camera is active and the user positions their face correctly, when the system captures the new facial image, then the system processes the capture and validates the new facial parameters.
- [ ] *AC3:* Given that the new facial parameters are valid, when the update is saved, then the system replaces the previous facial parameters with the new ones and reflects the changes in the system's behavior.
- [ ] *AC4:* Given that the facial capture fails due to lighting, movement, or camera issues, when the update attempt is processed, then the system shows a clear error message and allows the user to retry.
- [ ] *AC5:* Given that the user is not registered or has not logged into the system, when the user attempts to update facial parameters, then the system denies access and shows a clear authentication error message.
- [ ] *AC6:* Given that the facial parameter update is successful, when the system processes the change, then the system shows a confirmation message and records an audit event containing the user, date, and time.
- [ ] *AC7:* Given that the facial parameters are updated, when the user performs subsequent attendance registration, then the system uses the new facial parameters for recognition.
- [ ] *AC8:* Given that an administrator requests a facial parameter update on behalf of another user, when the administrator has appropriate permissions, then the system allows the update and records the administrator's identity in the audit event.
- [ ] *AC9:* Given normal device, camera, and network conditions, when the user completes the facial parameter update, then the system responds in less than 5 seconds, according to RNF 1.

Technical notes

- RF 4.4 indicates that the system must validate that the user is registered and has logged into the system before allowing facial parameter updates.

- RF 4.4 specifies that this operation can be requested by the user themselves or by an administrator.
- This story complements HU-ATT-002 (initial facial parameter enrollment). The enrollment story covers the first-time registration, while this story covers updates to existing parameters.
- The facial parameter update should follow the same security and data-protection requirements as the initial enrollment:
    - Biometric data must be stored securely.
    - Facial templates or embeddings should be preferred over raw images.
    - Compliance with Colombian Law 1581 of 2012 (Habeas Data).
- The system should consider:
    - Requiring explicit user confirmation before replacing existing parameters.
    - Maintaining a backup of the previous parameters for a defined retention period.
    - Invalidating the previous parameters immediately after the update.
- The facial parameter update may require re-authentication to confirm the user's identity before proceeding.
- The system must not expose facial parameters or biometric data in logs or interface messages.
- This story does not cover initial facial enrollment. That is handled in HU-ATT-002.
- This story does not cover user activation/deactivation. That is handled in HU-IAM-007 or HU-CONF-007 depending on the SRS interpretation.

Responsible service(s): Configuration service / Biometric service, proposed.

Endpoint(s) implemented:

- `POST /api/v1/users/me/facial-parameters/update` — proposed, for self-service updates.

- `POST /api/v1/users/{userId}/facial-parameters/update` — proposed, for administrator-initiated updates.

Events generated:

- `FacialParameterUpdated`
- `FacialParameterUpdateFailed`
- `UnauthorizedFacialParameterUpdateAttempt`

Required permissions:

- Authenticated user for self-service updates.
- Administrator or Supervisor for updates on behalf of other users.

Definition of Done (DoD)

This HU can only be closed when it meets the team's full DoD.

Additional checks specific to this HU:

- [ ] Successful self-service facial parameter update tested.
- [ ] Successful administrator-initiated update tested.
- [ ] Failed facial capture scenario tested.
- [ ] Unauthenticated access attempt tested.
- [ ] New parameters are used for subsequent attendance registration.
- [ ] Previous parameters are invalidated after update.
- [ ] User confirmation is required before replacing parameters.
- [ ] Biometric data is not exposed in logs or UI.
- [ ] Audit event is generated.
- [ ] Confirmation and error messages are clear.
- [ ] Response time is validated under normal conditions and remains under 5 seconds.
- [ ] Traceability to RF 4.4 is documented.

Estimation and priority

Table 28

Estimation and priority

Field	Value
Story Points	5
Priority	Medium
Target sprint	Sprint 2
Dependencies	HU-IAM-001: User login; HU-ATT-002: Initial facial parameter enrollment

Source. Own elaboration.

HU-CONF-007: IoT device failure alerts management

Story

As an administrator or supervisor

I want to receive alerts when IoT devices used for attendance registration experience failures

So that I can take timely corrective actions and maintain the availability of the attendance registration system.

Acceptance criteria

Format: "Given [context/initial state], when [user action], then [expected and verifiable result]."

- [ ] *AC1:* Given that an IoT device registered in the system detects a failure (camera malfunction, connectivity loss, hardware error), when the failure is detected, then the system generates an alert and notifies the configured administrators or supervisors.
- [ ] *AC2:* Given that an IoT device failure alert is generated, when the administrator accesses the alert management module, then the system displays the alert with device information, failure type, date, and time.
- [ ] *AC3:* Given that the administrator acknowledges the alert, when the acknowledgment is processed, then the system marks the alert as acknowledged and records the administrator's identity and timestamp.
- [ ] *AC4:* Given that the administrator resolves the device failure, when the resolution is recorded, then the system marks the alert as resolved and updates the device status

accordingly.

- [ ] *AC5:* Given that an IoT device reconnects or recovers automatically, when the system detects the recovery, then the system updates the device status and closes any open alerts associated with that device.
- [ ] *AC6:* Given that the user attempting to access the alert management module does not have administrator or supervisor permissions, when the request is processed, then the system denies access and records an unauthorized access attempt.
- [ ] *AC7:* Given normal network conditions, when the administrator accesses or manages IoT device alerts, then the system responds in less than 5 seconds, according to RNF 1.
- [ ] *AC8:* Given that an IoT device failure alert is generated, when the operation is completed, then the system records an audit event containing the device, failure type, date, and time.

Technical notes

- ⚠️ *SRS Inconsistency Note:* The SRS index lists RF 4.5 as "Gestión de alertas de fallos de dispositivos IoT" [IoT device failure alert management], but the detailed RF 4.5 section describes "Desactivar y reactivar usuarios" [Deactivate and reactivate users]. This story is documented based on the *index description* (IoT device failure alerts). The user activation/deactivation functionality is covered in HU-IAM-007 (RF 1.6). This inconsistency should be validated with the instructor.
- The IoT device failure alerts management should consider:
    - Device types: cameras, sensors, or other hardware used for attendance registration.
    - Failure types: connectivity loss, hardware malfunction, camera failure, low battery (if applicable).
    - Alert severity levels: informational, warning, critical.

- The alert notification mechanism should integrate with the notification service defined in HU-CONF-001 and HU-JUS-003.
- The system should consider:
    - Automatic health checks for registered IoT devices.
    - Configurable alert thresholds (e.g., alert after N consecutive failures).
    - Alert escalation for unresolved critical failures.
- The device status should be updated in real-time or near-real-time.
- The system should maintain a device registry with information such as device ID, location/environment, status, and last activity timestamp.
- This story does not cover user activation/deactivation. That is handled in HU-IAM-007.
- The team must define the specific IoT device types and failure detection mechanisms.

Responsible service(s): Configuration service / IoT monitoring service, proposed.

Endpoint(s) implemented:

- `GET /api/v1/iot-devices/alerts` — proposed, returns IoT device failure alerts.
- `PATCH /api/v1/iot-devices/alerts/{alertId}/acknowledge` — proposed, acknowledges an alert.
- `PATCH /api/v1/iot-devices/alerts/{alertId}/resolve` — proposed, resolves an alert.
- `GET /api/v1/iot-devices` — proposed, returns device registry with status.

Events generated:

- `IoTDeviceFailureDetected`
- `IoTDeviceAlertAcknowledged`
- `IoTDeviceAlertResolved`
- `IoTDeviceRecovered`
- `UnauthorizedIoTAlertAccessAttempt`

Required permissions:

- Administrator or Supervisor for alert management.
- Regular users, instructors, and students must not access IoT device alerts.

Definition of Done (DoD)

This HU can only be closed when it meets the team's full DoD.

Additional checks specific to this HU:

- [ ] IoT device failure detection tested.
- [ ] Alert generation tested for different failure types.
- [ ] Alert acknowledgment tested.
- [ ] Alert resolution tested.
- [ ] Device automatic recovery scenario tested.
- [ ] Unauthorized access attempt tested.
- [ ] Device registry displays correct status information.
- [ ] Audit event is generated for all alert operations.
- [ ] Confirmation and error messages are clear.
- [ ] Response time is validated under normal conditions and remains under 5 seconds.
- [ ] SRS inconsistency is documented and validated with instructor.
- [ ] Traceability to RF 4.5 is documented.

Estimation and priority

Table 29

Estimation and priority

Field	Value
Story Points	5
Priority	High
Target sprint	Sprint 3
Dependencies	HU-IAM-001: User login; HU-IAM-003: Role-based permission assignment; HU-CONF-001: Configure system alerts

Source. Own elaboration.

HU-CONF-008: Delay and justification parametrization

Story

As an administrator

I want to configure which excuses or justifications are valid for supporting absences or delays

So that the justification evaluation process follows institutional criteria and ensures consistent treatment of all student cases.

Acceptance criteria

Format: "Given [context/initial state], when [user action], then [expected and verifiable result]."

- [ ] *AC1:* Given that an authenticated administrator accesses the delay and justification parametrization module, when the system loads the module, then it displays the list of currently configured valid justifications and the option to add, modify, or remove parameters.
- [ ] *AC2:* Given that the administrator adds a new valid justification parameter, when the save operation is submitted, then the system validates that the parameter does not already exist in the system and stores the new justification type.
- [ ] *AC3:* Given that the justification parameter already exists in the system, when the administrator attempts to add it again, then the system rejects the operation and shows a clear error message indicating that the parameter is already defined.
- [ ] *AC4:* Given that the administrator modifies an existing justification parameter, when the changes are saved, then the system updates the parameter and reflects the changes in subsequent justification evaluations.
- [ ] *AC5:* Given that the administrator removes a justification parameter, when the removal is confirmed, then the system deactivates the parameter and it is no longer considered valid for new justification evaluations.
- [ ] *AC6:* Given that the user attempting to access the parametrization module does not have administrator permissions, when the request is processed, then the system denies

access and records an unauthorized access attempt.

- [ ] *AC7:* Given that a justification parameter is configured, when a student submits a justification in HU-JUS-001, then the system uses the configured parameters to determine if the justification is valid during evaluation in HU-JUS-002.
- [ ] *AC8:* Given normal network conditions, when the administrator saves the parametrization changes, then the system responds in less than 5 seconds, according to RNF 1.
- [ ] *AC9:* Given that a justification parameter is added, modified, or removed, when the operation is completed, then the system records an audit event containing the administrator, action type, parameter details, date, and time.

Technical notes

- RF 4.7 indicates that the system must validate that the configured parameter is not already defined in the system.
- RF 4.7 specifies that this is an *administrator-only* operation for defining valid justifications.
- The justification parametrization should consider:
    - Justification type name (e.g., medical certificate, family emergency, institutional activity).
    - Whether a supporting document is required.
    - Whether the justification applies to absences, delays, or both.
    - Maximum validity period (e.g., justification must be submitted within N days).
    - Whether the justification requires approval from a specific role.
- The configured parameters are used by:
    - HU-JUS-001: To validate that the submitted justification matches a valid type.
    - HU-JUS-002: To determine if the justification is valid during evaluation.

    - HU-REP-004: To filter and report delays and absences by justification status.
- The system should consider:
    - Preventing removal of justification parameters that are actively used in pending justifications.
    - Maintaining a historical record of parameter changes for audit purposes.
    - Providing default justification types for initial setup.
- The team must define whether justification parameters are institution-specific or global.
- This story does not cover the justification submission process. That is handled in HU-JUS-001.
- This story does not cover the justification evaluation process. That is handled in HU-JUS-002.
- This story does not cover delay detection. That is handled in HU-ATT-001.

Responsible service(s): Configuration service / Justification parametrization service, proposed.

Endpoint(s) implemented:

- `GET /api/v1/config/justifications` — proposed, returns configured justification parameters.
- `POST /api/v1/config/justifications` — proposed, adds a new justification parameter.
- `PUT /api/v1/config/justifications/{paramId}` — proposed, modifies an existing parameter.
- `DELETE /api/v1/config/justifications/{paramId}` — proposed, deactivates a parameter.

Events generated:

- `JustificationParameterAdded`

- `JustificationParameterModified`
- `JustificationParameterDeactivated`
- `JustificationParametrizationFailed`
- `UnauthorizedParametrizationAccessAttempt`

Required permissions:

- Administrator only.
- Regular users, instructors, and students must not access justification parametrization.

Definition of Done (DoD)

This HU can only be closed when it meets the team's full DoD.

Additional checks specific to this HU:

- [ ] Justification parametrization module is accessible only to administrators.
- [ ] Successful parameter addition tested.
- [ ] Duplicate parameter scenario tested.
- [ ] Successful parameter modification tested.
- [ ] Successful parameter deactivation tested.
- [ ] Unauthorized access attempt tested.
- [ ] Configured parameters are used in justification evaluation flow.
- [ ] Audit event is generated for all parametrization operations.
- [ ] Confirmation and error messages are clear.
- [ ] Response time is validated under normal conditions and remains under 5 seconds.
- [ ] Traceability to RF 4.7 is documented.

Estimation and priority

Table 30

Estimation and priority

Field	Value
Story Points	5
Priority	High

Field	Value
Target sprint	Sprint 2
Dependencies	HU-IAM-001: User login; HU-IAM-003: Role-based permission assignment

Source. Own elaboration.